Fundamentals of Statistical Learning - Unit 1

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Reference Bibliography

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Disclaimer

This document contains notes for the **Fundamentals of Statistical Learning - Unit 1** course, given by **Professor Brutti** as part of the Master's Degree in Data Science at **Sapienza Università di Roma** during the **first** semester of Academic Year **2026/2027**.

Note: These notes may not cover all details of each lecture, so it is highly suggested to consult the Professor's official resources and materials.

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Chapter 1

Defining probability

1.1 Introducing a qualitative idea of probability

Probability is the field of mathematics that studies events and how likely they are to occur in order to describe relationships or patterns in existing models.
Starting from a model's data generation process, probability aims to investigate the properties of the observed outcomes, whereas inference makes use of the observed outcomes to study the underlying data generation process.

Example: Diffusion models

Diffusion models are a type of generative model that create new data by gradually reversing a noising process.
From a mathematical perspective, a diffusion model can be described using a Markovian directed acyclic graph whose nodes represent the observable or latent variables involved in the process and whose edges denote the transition probabilities of the model.

1.2 Formalizing probability using set theory

It is possible to formalize probabilistic experiments using set theory.
In fact, starting from a sample space Ω, which represents the set of all possible outcomes, an event A on Ω can be defined as the set of all favourable outcomes in which the event does occur.

Example: Tossing a coin

Suppose you toss a coin: it is possible to construct a random experiment by defining the sample space $\Omega = \{Heads, Tails\}$ and considering the event $A = \{Heads\}$.

1.2.1 Set operations on probability spaces

Given two events A and B on the same sample space Ω, it is possible to define three main operations:

- **Union:** The set $A \cup B = \{\omega \in \Omega : \omega \in A \text{ or } \omega \in B\}$ corresponds to the event in which at least one of A or B, or both, occurs.
  In particular, it is always the case that $A \cup \Omega = \Omega$ and $A \cup \emptyset = A$.
- **Intersection:** The set $A \cap B = \{\omega \in \Omega : \omega \in A \text{ and } \omega \in B\}$ corresponds to the event in which both A and B occur.
  In particular, it is always the case that $A \cap \Omega = A$ and $A \cap \emptyset = \emptyset$.
- **Complement:** The set $A^C = \Omega \setminus A = \{\omega \in \Omega : \omega \notin A\}$ corresponds to the event in which A does not occur.
  In particular, it is always the case that $A \cup A^C = \Omega$ and $A \cap A^C = \emptyset$.

Additionally, given a sequence of events $\{A_1, \ldots, A_n\}$ on the same sample space Ω, De Morgan's laws allow to switch between the union and intersection operations through the following identities:

$$\left(\bigcup_{i=1}^{n} A_i\right)^C = \bigcap_{i=1}^{n} A_i^C \text{ and } \left(\bigcap_{i=1}^{n} A_i\right)^C = \bigcup_{i=1}^{n} A_i^C$$

1.2.2 Sample space partitions

A sequence of events $\{A_1, \ldots, A_n\}$ is said to form a partition on a sample space Ω if the following conditions are satisfied:

1. $A_1 \cup \cdots \cup A_n = \Omega$, typically requiring $A_i \neq \emptyset \ \forall i = 1, \ldots, n$ as well.
2. The sets in the sequence are pairwise disjoint, meaning that $A_i \cap A_j = \emptyset \ \forall i \neq j$.

Generally speaking, partitions come in handy to explain the probability of an event by decomposing it as a union of disjoint sets with respect to a sequence of mutually exclusive hypotheses.

1.3 Probability measure functions

While several definitions of probability exist, Kolmogorov's axiomatic definition is the most commonly used when defining probability measures.
According to this definition, a probability measure function $\mathbb{P} : \mathcal{P}(\Omega) \to [0, 1]$ assigns to each event $A \subseteq \Omega$ a probability $\mathbb{P}(A)$ such that:

1. $\mathbb{P}(A) \in [0, 1] \ \forall A \subseteq \Omega$.
2. $\mathbb{P}(\emptyset) = 0$ and $\mathbb{P}(\Omega) = 1$.
3. $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$, with $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$ if and only if $A \cap B = \emptyset$.
   This last axiom can be generalized for a sequence of n events through the inclusion-exclusion principle, which states that

$$\mathbb{P}\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{k=1}^{n} (-1)^{k+1} \left(\sum_{1 \leq i_1 < \cdots < i_k \leq n} \mathbb{P}(A_{i_1} \cap \ldots A_{i_k})\right)$$

Example: Checking polynomial identities using a randomized algorithm

Consider two polynomials of degree d:

$$F(x) = \prod_{i=1}^{d}(x - a_i) \text{ and } G(x) = \sum_{i=1}^{d} c_i x^i$$

A simple way of checking whether $F(x) \equiv G(x)$ involves expanding $F(x)$ into its canonical form and compare the resulting coefficient with those of $G(x)$, although this algorithm would require performing $\Theta(d^2)$ operations.
Knowing that, for any randomly chosen point x_0, evaluating $F(x_0)$ and $G(x_0)$ requires $O(d)$ time, it is possible to cut down complexity using a randomized algorithm:

Algorithm 1: Randomized Algorithm for Polynomial Identities.

Data: $F(x), G(x)$
Result: Whether $F(x) \equiv G(x)$.
Generate r uniformly at random from a set $\mathcal{I} = \{1, \dots, 100d\}$;
Compute $F(r)$ and $G(r)$;
if $F(r) = G(r)$ **then**
  $F(x) \equiv G(x)$;
else
  $F(x) \not\equiv G(x)$;

However, the randomized nature of this algorithm can result in a wrong evaluation.
In fact, while $F(x) \equiv G(x)$ results in $F(r) = G(r)\ \forall r$, it might be the case that, for some r, $F(r) = G(r)$ even if $F(x) \not\equiv G(x)$: this implies that the algorithm has one-sided error as it can make mistakes only when deciding that $F(x) \equiv G(x)$.
Therefore, define the event

$$E = \{F(r) = G(r) \text{ but } F(x) \not\equiv G(x)\}$$

This event occurs if and only if r is a root of the polynomial $H(x) = F(x) - G(x)$: since $H(x)$ will always have degree at most d, it will have at most d roots, allowing to bound the error probability as follows:

$$\mathbb{P}(E) = \frac{|\{r \in \mathcal{I} : r \text{ is a root of } H(x)\}|}{|\mathcal{I}|} \leq \frac{d}{100d} = \frac{1}{100}$$

As a consequence to Kolmogorov's axiomatic definition of probability, every probability space $(\Omega, \mathbb{P})$ satisfies the property that $\mathbb{P}(A^C) = 1 - \mathbb{P}(A)\ \forall A \subseteq \Omega$.

Proof:

Consider the partition $\Omega = A \cup A^C$ and apply Kolmogorov's axioms.

$$1 = \mathbb{P}(\Omega) = \mathbb{P}(A \cup A^C) = \mathbb{P}(A) + \mathbb{P}(A^C) \Rightarrow \mathbb{P}(A^C) = 1 - \mathbb{P}(A)$$

1.3.1 Probability bounds

Boole's inequality states that, for every sequence of events $\{A_1, \ldots, A_n\}$, it holds that

$$\mathbb{P}\left(\bigcup_{i=1}^{n} A_i\right) \leq \sum_{i=1}^{n} \mathbb{P}(A_i)$$

Proof:

This inequality can be proved by induction on the length of the sequence.

- For the base case, give by $n = 1$, the inequality is trivially true as $\mathbb{P}(A_1) \leq \mathbb{P}(A_1)$.
- Assuming the inequality is true for some n, the goal is to show that it is true for $(n+1)$ as well.
  Reformulate $A_1 \cup \cdots \cup A_{n+1} = (A_1 \cup \cdots \cup A_n) \cup A_{n+1}$ and let

$$\mathbb{P}\left(\bigcup_{i=1}^{n+1} A_i\right) = \mathbb{P}\left(\left(\bigcup_{i=1}^{n} A_i\right) \cup A_{n+1}\right) = \mathbb{P}\left(\bigcup_{i=1}^{n} A_i\right) + \mathbb{P}(A_{n+1}) - \mathbb{P}\left(\left(\bigcup_{i=1}^{n} A_i\right) \cap A_{n+1}\right)$$
$$\overset{\mathbb{P}(.)\geq 0}{\leq} \mathbb{P}\left(\bigcup_{i=1}^{n} A_i\right) + \mathbb{P}(A_{n+1}) \overset{\text{hp. ind.}}{\leq} \sum_{i=1}^{n} \mathbb{P}(A_i) + \mathbb{P}(A_{n+1}) = \sum_{i=1}^{n+1} \mathbb{P}(A_i)$$

Having proved the base case and the inductive case, this inequality must be true $\forall n \in \mathbb{N}$, which implies that probability measure functions are always sub-additive.

As a consequence to Boole's inequality, Bonferroni's inequality states that, for every sequence of events $\{A_1, \ldots, A_n\}$, it must be the case that

$$\mathbb{P}\left(\bigcap_{i=1}^{n} A_i\right) \geq \sum_{i=1}^{n} \mathbb{P}(A_i) - (n-1)$$

Proof:

Reformulate the joint probability of the events using De Mrogan's laws.

$$\mathbb{P}\left(\bigcap_{i=1}^{n} A_i\right) = 1 - \mathbb{P}\left(\left(\bigcap_{i=1}^{n} A_i\right)^C\right) \overset{\text{De Morgan}}{=} 1 - \mathbb{P}\left(\bigcup_{i=1}^{n} A_i^C\right)$$
$$\overset{\text{Boole}}{\geq} 1 - \sum_{i=1}^{n} \mathbb{P}(A_i^C) = 1 - \sum_{i=1}^{n} (1 - \mathbb{P}(A_i))$$
$$= 1 - \left(\sum_{i=1}^{n} 1 - \sum_{i=1}^{n} \mathbb{P}(A_i)\right) = \sum_{i=1}^{n} \mathbb{P}(A_i) - (n-1)$$

Chapter 2

Conditional probability

2.1 Understanding conditional probability

Given a sample space Ω, the conditional probability that event A occurs knowing that event B was observed can be defined as follows:

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}, \text{ with } \mathbb{P}(B) > 0.$$

Example: Measles and red spots

Consider the two events

$$A = \{\text{I have measles}\} \text{ and } B = \{\text{I have red spots on my face}\}$$

Knowing that red spots are a common symptom of measles, it will be the case that $\mathbb{P}(B|A) = 1$. However, the same thing cannot be said about $\mathbb{P}(A|B)$ as red spots might be caused by other external factors, such as eating hot food or getting a scratch.

From a broader perspective, a conditional probability distribution can be described using a probability measure function that satisfies the following properties:

1. $\mathbb{P}(A|B) \in [0, 1] \ \forall A \subseteq \Omega$.
2. $\mathbb{P}(\Omega|B) = 1$.
3. For any sequence of disjoint events $\{A_1, \dots A_n\}$, it is always the case that

$$\mathbb{P}\left(\bigcup_{i=1}^{n} A_i \middle| B\right) = \sum_{i=1}^{n} \mathbb{P}(A_i|B)$$

As a consequence, it is always the case that $\mathbb{P}(A^C|B) = 1 - \mathbb{P}(A|B) \ \forall A \subseteq \Omega$.

Proof:

Let $B = B \cap \Omega$ on the partition $\Omega = A \cup A^C$.

$$\begin{aligned}\mathbb{P}(B) = \mathbb{P}(B \cap \Omega) = \mathbb{P}(B \cap (A \cup A^C)) = \mathbb{P}((B \cap A) \cup (B \cap A^C)) \\ = \mathbb{P}(A \cap B) + \mathbb{P}(A^C \cap B) \Rightarrow \mathbb{P}(A^C \cap B) = \mathbb{P}(B) - \mathbb{P}(A \cap B)\end{aligned}$$

Therefore, applying the definition of conditional probability will result in

$$\mathbb{P}(A^C|B) = \frac{\mathbb{P}(A^C \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(B) - \mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(B)}{\mathbb{P}(B)} - \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = 1 - \mathbb{P}(A|B)$$

2.1.1 The (conditional) odds of an event

Generally speaking, the odds of an event can be seen as the likelihood that the event will occur relative to it not occurring.
For this reason, the odds of an event $A \subseteq \Omega$ with probability $\mathbb{P}(A) \in [0, 1)$ can be computed as

$$\mathcal{O}(A) = \frac{\mathbb{P}(A)}{\mathbb{P}(A^C)} = \frac{\mathbb{P}(A)}{1 - \mathbb{P}(A)}$$

More specifically, the conditional odds of event A given that a second event B occurred can be computed as

$$\mathcal{O}(A|B) = \frac{\mathbb{P}(A|B)}{\mathbb{P}(A^C|B)} = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A^C \cap B)}$$

2.1.2 The multiplication rule

The multiplication rule allows to reformulate the joint probability of two events A and B according to the following equation:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A|B)\mathbb{P}(B) = \mathbb{P}(B|A)\mathbb{P}(A)$$

In particular, repeatedly applying the multiplication rule to a sequence of events $\{A_1, \ldots, A_n\}$ results in the chain rule, which can be formulated as

$$\mathbb{P}\left(\bigcap_{i=1}^{n} A_i\right) = \mathbb{P}(A_1) \cdot \mathbb{P}(A_2|A_1) \cdot \cdots \cdot \mathbb{P}(A_n|A_1 \cap \ldots A_{n+1})$$

2.2 Defining event independence

Two events A and B on a sample space Ω are said to be independent if knowing that B occurred does not add information about the probability of A occurring, resulting in

$$A \perp B \Leftrightarrow \mathbb{P}(A|B) = \mathbb{P}(A) \text{ and } \mathbb{P}(B|A) = \mathbb{P}(B), \text{ resulting in } \mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

Depending on the context, event independence can be assumed or verified, although it is always possible to show that, if A and B are events such that $\mathbb{P}(A), \mathbb{P}(B) > 0$ and $A \cap B = \emptyset$, then the events cannot be independent.

Proof:

It is possible to notice that having $A \cap B = \emptyset$ and knowing that one of the two events occurred would violate the definition of event independence because

$$\mathbb{P}(A)\mathbb{P}(B) > 0, \text{ but } \mathbb{P}(A \cap B) = \mathbb{P}(\emptyset) = 0.$$

2.3 Generalizing event independence

For $n > 2$ events, independence can be defined at several distinct levels.

2.3.1 Pairwise and mutual independence

A (countable) sequence of events $\{A_1, \ldots, A_n\}$ is said to be pairwise independent if

$$A_i \perp A_j \ \forall i \neq j \Leftrightarrow \mathbb{P}(A_i \cap A_j) = \mathbb{P}(A_i)\mathbb{P}(A_j) \ \forall i \neq j$$

Additionally, a (countable) sequence of events $\{A_1, \ldots, A_n\}$ is said to be mutually independent if, for any group of k events in the sequence, it is the case that

$$\mathbb{P}\left(\bigcap_{i=1}^{k} A_{n_i}\right) = \prod_{i=1}^{k} \mathbb{P}(A_{n_i}), \text{ for } 1 \leq k \leq n \text{ and } k \in \mathbb{N}.$$

It should be noticed that, while mutual independence always implies pairwise independence, the converse statement is not true because a sequence of events can be pairwise independent but not mutually independent.

Example: A special die

Given a balanced four-faced die, which has a red face, a green face, a blue face and a striped face with all three colours, define the events

$$R = \{\text{I see red}\}, \ G = \{\text{I see green}\} \text{ and } B = \{\text{I see blue}\}$$

It is possible to show that these events are pairwise independent as

$$\begin{aligned}
\mathbb{P}(R \cap G) &= 0.25 = 0.5 \cdot 0.5 = \mathbb{P}(R)\mathbb{P}(G) \\
\mathbb{P}(R \cap B) &= 0.25 = 0.5 \cdot 0.5 = \mathbb{P}(R)\mathbb{P}(B) \\
\mathbb{P}(G \cap B) &= 0.25 = 0.5 \cdot 0.5 = \mathbb{P}(G)\mathbb{P}(B)
\end{aligned}$$

However, the events are not mutually independent because

$$\mathbb{P}(R \cap G \cap B) = 0.25 \neq 0.125 = 0.5 \cdot 0.5 \cdot 0.5 = \mathbb{P}(R)\mathbb{P}(G)\mathbb{P}(B)$$

2.3.2 Conditional independence

Two events A and B are said to be conditionally independent given an event C if, knowing that C occurred, observing event B does not add information about the probability of A occurring, resulting in

$$\mathbb{P}(A \cap B|C) = \mathbb{P}(A|C)\mathbb{P}(B|C), \text{ with } \mathbb{P}(C) > 0.$$

Example: University admissions

For a student applying to a Master's Degree course, define the events

$$DS = \{\text{Accepted to Data Science}\} \text{ and } ST = \{\text{Accepted to Statistics}\}$$

Knowing that event DS occurred can provide additional information about the likelihood of event ST occurring, meaning that the two events are not independent.
However, if the admission procedure is solely based on the student's GPA C, knowing that one event occurred does not provide additional information about the other event occurring as well, making DS and ST conditionally independent given event C.

2.4 The law of total probability

Given a sample space Ω, the law of total probability allows to explain the probability of an event $A \subseteq \Omega$ by decomposing it over the partition $\Omega = B \cup B^C$, resulting in

$$\mathbb{P}(A) = \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^C) = \mathbb{P}(A|B)\mathbb{P}(B) + \mathbb{P}(A|B^C)\mathbb{P}(B^C)$$

Proof:

Let $A = A \cap \Omega$ on the partition $\Omega = B \cup B^C$.

$$\begin{aligned}\mathbb{P}(A) &= \mathbb{P}(A \cap \Omega) = \mathbb{P}(A \cap (B \cup B^C)) = \mathbb{P}((A \cap B) \cup (A \cap B^C)) \\ &= \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^C) \overset{\text{Multiplication rule}}{=} \mathbb{P}(A|B)\mathbb{P}(B) + \mathbb{P}(A|B^C)\mathbb{P}(B^C)\end{aligned}$$

Additionally, it is possible to generalize the law of total probability to explain the probability of an event E with respect to a sequence of mutually exclusive hypotheses $\{H_1, \ldots, H_n\}$ as

$$\mathbb{P}(E) = \sum_{i=1}^{n} \mathbb{P}(E \cap H_i) = \sum_{i=1}^{n} \mathbb{P}(E|H_i)\mathbb{P}(H_i)$$

2.5 Bayes' theorem

Bayes' theorem allows to switch between conditional probabilities by setting

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(B|A)\mathbb{P}(A)}{\mathbb{P}(B)} \text{ with } \mathbb{P}(B) > 0.$$

In particular, when considering a sequence of mutually exclusive hypotheses $\{H_1, \ldots, H_n\}$ with known prior probabilities, Bayes' theorem allows to compute the posterior probability of hypothesis H_i given the observed evidence E as

$$\mathbb{P}(H_i|E) = \frac{\mathbb{P}(E|H_i)\mathbb{P}(H_i)}{\mathbb{P}(E)} = \frac{\mathbb{P}(E|H_i)\mathbb{P}(H_i)}{\sum_{j=1}^{n} \mathbb{P}(E|H_j)\mathbb{P}(H_j)}$$

In fact, the posterior probability of each hypothesis given the observed evidence will be computed in terms of the likelihood $\mathbb{P}(E|H_i)$, which describes the probability of seeing the evidence given that the hypothesis H_i is true, the prior probability $\mathbb{P}(H_i)$, which describes the probability of H_i being true, and the marginal probability $\mathbb{P}(E)$, which denotes the total probability of seeing the evidence under all possible hypotheses.

Example: Iterated Bayes for medical diagnosis

Consider the events

$$P = \{\text{I test positive to a disease}\} \text{ and } D = \{\text{I actually have the disease}\}$$

In particular, the disease affects 0.1% of the population and it is known that a dedicated test correctly detects 99% of disease case but produces a false positive result on 1% of occasions, which means that

$$\mathbb{P}(D|P) = \frac{\mathbb{P}(P|D)\mathbb{P}(D)}{\mathbb{P}(P|D)\mathbb{P}(D) + \mathbb{P}(P|D^C)\mathbb{P}(D^C)} = \frac{0.99 \cdot 0.001}{(0.99 \cdot 0.001) + (0.01 \cdot 0.999)} = 9\%$$

Since the posterior probability appears to be quite low, Bayes' rule is applied once again, using the posterior probability as the new prior probability, resulting in

$$\mathbb{P}(D|P) = \frac{\mathbb{P}(P|D)\mathbb{P}(D)}{\mathbb{P}(P|D)\mathbb{P}(D) + \mathbb{P}(P|D^C)\mathbb{P}(D^C)} = \frac{0.99 \cdot 0.09}{(0.99 \cdot 0.09) + (0.01 \cdot 0.91)} = 91\%$$

2.5.1 The posterior odds of a hypothesis

When considering an event E on a sequence of mutually exclusive hypotheses $\{H_1, \ldots, H_n\}$, it is possible to compute the posterior odds of hypothesis H_i given evidence E as

$$\mathcal{O}(H_i|E) = \frac{\mathbb{P}(H_i|E)}{\mathbb{P}(H_i^C|E)} = \frac{\frac{\mathbb{P}(E|H_i)\mathbb{P}(H_i)}{\mathbb{P}(E)}}{\frac{\mathbb{P}(E|H_i^C)\mathbb{P}(H_i^C)}{\mathbb{P}(E)}} = \frac{\mathbb{P}(E|H_i)\mathcal{O}(H_i)}{\mathbb{P}(E|H_i)}$$

Chapter 3

Random variables

3.1 Defining random variables

Given a probability space $(\Omega, \mathbb{P})$, a random variable is a (measurable) mapping $X : \Omega \to S_X \subseteq \mathbb{R}$ that maps every outcome $\omega \in \Omega$ onto a value $X(\omega) \in \mathbb{R}$, with the set set $S_X = \{X(\omega) : \omega \in \Omega\}$ forming the "support" of X.
This concept can be extended to define a random vector, also known as a p-variate random variable, as a (measurable) mapping $\mathbf{X} : \Omega \to \mathbb{R}^p$ that assigns a vector $\mathbf{X}(\omega) \in \mathbb{R}^p$ to each outcome $\omega \in \Omega$.

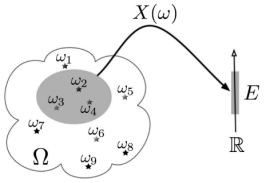


Figure 3.1: A visual idea of what is a random variable.

Since this mapping describes the probability of the random variable X falling within a subset of $\mathbb{R}$, it is possible to define the pre-image of a set $E \subseteq \mathbb{R}$ as the set $X^{-1}(E) = \{\omega \in \Omega : X(\omega) \in E\} \subseteq \Omega$. This definition allows to describe the probability distribution of a random variable X as

$$\mathbb{P}(X \in E) = \mathbb{P}(\{\omega \in \Omega : X(\omega) \in E\}) = \mathbb{P}(X^{-1}(E))$$

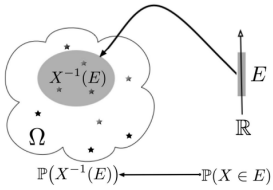


Figure 3.2: A visual idea of the pre-image of a random variable.

When defining the probability distribution of a random variable X, it should be noticed that $\mathbb{P}(X \in E)$ represents an induced distribution supported on $\mathbb{R}$ that is constructed based on the assumed distribution $\mathbb{P}(X^{-1}(E))$ supported on Ω.

3.1.1 Discrete random variables

A random variable is said to be "discrete" if its support is a countable set, which means that the random variable can take at most a finite, or at least countable, number of values.
For this reason, a discrete random variable $X : \Omega \to S_X$ can be described using a probability mass function $p_X(x) = \mathbb{P}(X = x)$ that satisfies the following conditions:

- $p_X(x) \geq 0 \ \forall x \in \mathbb{R}$, with $p_X(x) > 0 \ \forall x \in S_X$ specifically.
- $\sum_{x \in S_X} p_X(x) = 1.$

Example: Repeated coin flips

Consider 5 independent flips of a fair coin: since each flip can be seen as a repetition of a binary experiment with sample space $\Omega = \{Heads, Tails\}$, it is possible to define the overall outcome space as the Cartesian product $\Omega^{\otimes 5}$.
Assuming symmetry and independence between coin flips, each outcome $\omega \in \Omega^{\otimes 5}$ is equally likely and has probability $\mathbb{P}(\omega) = \dfrac{1}{|\Omega^{\otimes 5}|} = \dfrac{1}{32}$.
Define now the random variable $X = \{\text{Number of tails in 5 independent coin flips}\}$: on its support $S_X = \{0, 1, 2, 3, 4, 5\}$, the probability definition of X can be defined as

$$p_X(x) = \mathbb{P}(X = x) = \binom{5}{x} \left(\frac{1}{2}\right)^5, \text{ for } x \in S_X.$$

3.1.2 Continuous random variables

A random variable is said to be "continuous" if its support is an uncountable set.
Since, in this setting, $\mathbb{P}(X = x) \to 0 \ \forall x \in \mathbb{R}$, a continuous random variable $X : \Omega \to S_X$ is described using a probability density function $f_X(x)$ that satisfies the following conditions:

- $\mathbb{P}(a \leq X \leq b) = \mathbb{P}(X \in [a, b]) = \int_a^b f_X(x) \ dx.$
- $f_X(x) \geq 0 \ \forall x \in \mathbb{R}$, with $f_X(x) > 0 \ \forall x \in S_X$ specifically.
- $\int_{\mathbb{R}} f_X(x) = 1.$

In particular, all information about a continuous random variable X can be retrieved from its cumulative distribution function $F_X : \mathbb{R} \to [0, 1]$, which can be defined as

$$F_X(x) = \mathbb{P}(X \leq x) = \mathbb{P}(X \in (-\infty, x]) \ \forall x \in \mathbb{R}$$

More specifically, a function $F_X : \mathbb{R} \to [0, 1]$ acts as a cumulative distribution function on some probability measure $\mathbb{P}$ if and only if the following conditions are met:

- $F_X(x)$ is non-decreasing, which means that $F_X(x_1) \leq F_X(x_2)\ \forall x_1 < x_2$.
- $F_X(x)$ is normalized, meaning that

$$\lim_{x\to-\infty} F_X(x) = 0 \qquad \lim_{x\to+\infty} F_X(x) = 1$$

- $F_X(x)$ is right-continuous, which means that

$$F_X(x) = F_X(x^+) = \lim_{t\to x^+} F_X(t)$$

Furthermore, the cumulative distribution function of a random variable X always satisfies several properties:

- $\mathbb{P}(x_1 < X \leq x_2) = \mathbb{P}(X \in (x_1, x_2]) = F(x_2) - F(x_1)$.
- $\mathbb{P}(X > x) = \mathbb{P}(X \in (x, +\infty)) = 1 - F_X(x)$.
- If X is a continuous random variable with probability density function $f_X(x)$, then

$$F_X(x) = \int_{-\infty}^{x} f_X(t)\ dt \text{ and } f_X(x) = F'(x) \text{ wherever } F_X \text{ is differentiable.}$$

  It should be noticed that this last property can also be formulated for discrete random variables. In fact, if X is a discrete random variable with probability mass function $p_X(x)$, then its cumulative function is given by

$$F_X(x) = \sum_{x_i \leq x} p_X(x_i) = \mathbb{P}(X < x) + p_X(x), \text{ with } p_X(x) = F_X(x) - F_X(x^-).$$

Example: Playing darts

Suppose you throw a dart on a disc of radius r at random.
Define the random variable $X = \{\text{Distance of the hitting point from the center}\}$ and note that its support $[0, r]$ is an uncountable set.

$$\mathbb{P}(X \leq x) = 0\ \forall x < 0 \text{ as distances are not negative.}$$
$$\mathbb{P}(X \leq x) = 1\ \forall x > r \text{ as the dart did hit the disc.}$$
$$\mathbb{P}(X \leq x) = \frac{\pi x^2}{\pi r^2} = \frac{x^2}{r^2}\ \forall x \in [0, r]$$

Therefore, it is possible to define the cumulative distribution function of X as

$$F_X(x) = \begin{cases} 0 & \text{if } x < 0 \\ \frac{x^2}{r^2} & \text{if } x \in [0, r] \\ 1 & \text{if } x > r \end{cases}$$

This implies that the probability density function will be

$$f_X(x) = \begin{cases} \frac{2x}{r^2} & \text{if } x \in [0, r] \\ 0 & \text{otherwise} \end{cases}, \text{ or } f_X(x) = \frac{2x}{r^2}\mathbb{I}_{[0,r]}(x), \text{ with } \mathbb{I}_{[0,r]}(x) = \begin{cases} 1 & \text{if } x \in [0, r] \\ 0 & \text{otherwise} \end{cases}$$

3.2 The quantile function of a random variable

Let $X : \Omega \to S_X$ be a continuous random variable with cumulative distribution function $F_X(x)$: for a given level $p \in [0, 1]$, the quantile function is defined as

$$q_p = \mathbb{Q}(p) = \inf\{x \in S_X : F_X(x) = p\}$$

In particular, if $F_X(x)$ is strictly increasing in $\mathbb{R}$, then it is also invertible and it is possible to set

$$q_p = \mathbb{Q}(p) = F_X^{-1}(p) \text{ and } p = F_X(q_p) = \int_{-\infty}^{q_p} f_X(t)\ dt.$$

Example: The median of a Pareto distribution

A continuous random variable X is said to follow a Pareto distribution with parameter $\alpha > 0$ if its cumulative distribution function is defined as

$$F_X(x) = \begin{cases} 0 & \text{if } x < 1 \\ 1 - \frac{1}{x^\alpha} & \text{if } x \geq 1 \end{cases}$$

Assuming $\alpha = 0.5$, suppose you want to find the median of the distribution: since $F_X(x)$ is strictly increasing in $[1, +\infty)$, it is possible to find

$$F(q_p) = 1 - \frac{1}{\sqrt{q_p}} = \frac{\sqrt{q_p} - 1}{\sqrt{q_p}} = 0.5 \Rightarrow 0.5\sqrt{q_p} = \sqrt{q_p} - 1 \Rightarrow 0.5\sqrt{q_p} = 1 \Rightarrow q_p = 4$$

However, since discrete random variables tend to have weakly monotonic cumulative distribution functions, a more general definition of quantile function is typically used.
In fact, given any random variable $X : \Omega \to S_X$ with cumulative distribution function $F_X(x)$, the quantile function for a given level $p \in [0, 1]$ is formulated as

$$q_p = \mathbb{Q}(p) = \inf\{x \in S_X : F_X(x) \geq p\}$$

Example: The median of a discrete distribution

Let X be a discrete random variable whose probability distribution is given by

$$p_X(x) = \begin{cases} \frac{2}{5} & \text{if } x = -1 \\ \frac{1}{5} & \text{if } x = 0 \\ \frac{2}{5} & \text{if } x = 1 \end{cases} \qquad F_X(x) = \begin{cases} 0 & \text{if } x < -1 \\ \frac{2}{5} & \text{if } x \in [-1, 0) \\ \frac{3}{5} & \text{if } x \in [0, 1) \\ 1 & \text{if } x \geq 1 \end{cases}$$

Suppose you want to find the median of the distribution: since $F_X(x)$ is not strictly increasing, it is necessary to use the general definition of quantile function to find

$$q_p = \inf\{x \in S_X : F_X(x) \geq 0.5\} = 0$$

3.3 Common univariate distributions

Sometimes, the behaviour of a random variable can be described by some known theoretical probability distribution.

3.3.1 The Bernoulli distribution

A discrete random variable X is said to follow a Bernoulli distribution with parameter $p \in [0, 1]$, which is written as $X \sim Ber(p)$, if its probability mass function is defined as

$$p_X(x) = p^x(1-p)^{1-x}, \text{ for } x \in \{0, 1\}.$$

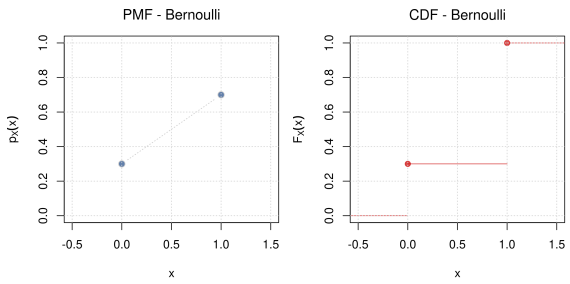


Figure 3.3: PMF and CDF for a random variable $X \sim Ber(0.7)$.

For this reason, Bernoulli random variables come in handy to model binary experiments whose only possible outcomes are success, which is denoted by $X = 1$, and failure, which is denoted by $X = 0$.

Example: Logistic regression

Logistic regression is a classification model that aims to predict a binary label y based on a feature vector $\mathbf{x}$.
For this reason, it is possible to model a conditional probability $\mathbb{P}(y|\mathbf{x}) \sim Ber(p(\mathbf{x}))$ whose success probability is computed using the logistic function

$$\ln\left(\frac{p(\mathbf{x})}{1-p(\mathbf{x})}\right) = \alpha + \beta^T\mathbf{x} \Rightarrow p(\mathbf{x}) = \frac{e^{\alpha+\beta^T\mathbf{x}}}{1+e^{\alpha+\beta^T\mathbf{x}}}$$

Moreover, if X is a random variable with probability density function $f_X(x)$, or probability mass function $p_X(x)$, then, for some event A, it holds that $Y = r(X) = \mathbb{I}_A(X) \sim Ber(p)$, with

$$p = \int_A f_X(x)\ dx$$

3.3.2 The binomial distribution

A discrete random variable X is said to follow a binomial distribution with parameters $n \in \mathbb{N}_0$ and $p \in [0, 1]$, which is written as $X \sim Bin(n, p)$, if its probability mass function is defined as

$$p_X(x) = \binom{n}{x} p^x (1-p)^{n-x}, \text{ for } x \in \{0, \dots, n\}.$$

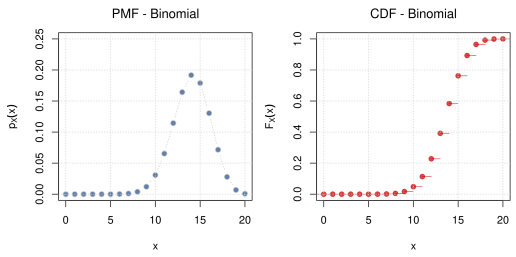


Figure 3.4: PMF and CDF for a random variable $X \sim Bin(20, 0.7)$.

Since binomial random variables are often used to model experiments counting the number of successful attempts out of n independent and identically distributed Bernoulli trials with success probability p, it is also possible to define a random variable $X \sim Bin(n, p)$ as

$$X = \sum_{i=1}^{n} X_i, \text{ with } X_i \sim Ber(p) \ \forall i \in \{1, \dots, n\}.$$

Example: Guessing card symbols

A man who claims to be a psychic is asked to guess the symbols drawn on 25 cards. Since the man guesses at random with success probability $p = 0.7$ at each round, define the random variable $X_i = \{\text{The } i^{th} \text{ guess is correct}\}$ and note that $X_i \sim Ber(p) \ \forall i \in \{1, \dots, 25\}$. For this reason, the random variable $X = \{\text{Total correct guesses}\}$ can be formulated as

$$X = \sum_{i=1}^{25} X_i \Rightarrow X \sim Bin(25, 0.7) \Rightarrow p_X(x) = \binom{25}{x} (0.7)^x (0.3)^{25-x}, \text{ for } x \in \{0, \dots, 25\}.$$

This means that the probability that the man correctly guesses each symbol will be

$$p_X(25) = \binom{25}{25} (0.7)^{25} (0.3)^{25-25} = (0.7)^{25} \approx 1.34 \cdot 10^{-4}$$

3.3.3 The geometric distribution

A discrete random variable X is said to follow a geometric distribution with parameter $p \in (0, 1)$, which is written as $X \sim Geom(p)$, if its probability mass function is defined as

$$p_X(x) = (1-p)^{x-1} p, \text{ for } x \in \mathbb{N}.$$

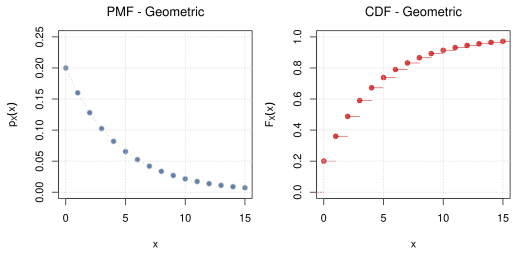


Figure 3.5: PMF and CDF for a random variable $X \sim Geom(0.2)$.

This distribution is typically used to model the number of trials, or failures, until the first successful attempt of a Bernoulli experiment because, due to its memoryless behaviour, the previous failures do not influence future outcomes, which means that

$$\mathbb{P}(X > x + k | X > x) = \mathbb{P}(X > k) \ \forall k > 0$$

Proof:

If $X \sim Geom(p)$, then it is possible to observe that

$$\mathbb{P}(X > n) = \sum_{k=n+1}^{+\infty} \mathbb{P}(X = k) = (1-p)^n \left(\sum_{i=1}^{+\infty} (1-p)^{i-1} p \right) \overset{\sum_{i=1}^{+\infty}(1-p)^{i-1} = \frac{1}{p}}{=} (1-p)^n$$

Therefore, by applying the definition of conditional probability, it is possible to conclude that

$$\mathbb{P}(X > x + k | X > x) = \frac{\mathbb{P}(\{X > x + k\} \cap \{X > x\})}{\mathbb{P}(X > x)} = \frac{(1-p)^{x+k}}{(1-p)^x} = (1-p)^k = \mathbb{P}(X > k)$$

3.3.4 The Poisson distribution

A discrete random variable X is said to follow a Poisson distribution with parameter $\mu \in \mathbb{R}^+$, which is written as $X \sim Pois(\mu)$, if its probability mass function is defined as

$$p_X(x) = \frac{e^{-\mu} \mu^x}{x!}, \text{ for } x \in \mathbb{N}_0.$$

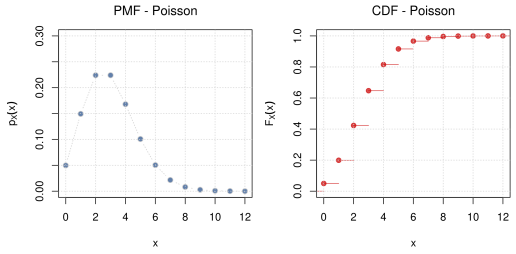


Figure 3.6: PMF and CDF for a random variable $X \sim Pois(3)$.

Since this distribution is typically used to analyse the distribution of rare events within a short amount of time, a random variable $X \sim Binom(n, p)$ can be approximated using a Poisson distribution with parameter $\mu = np$ whenever n is sufficiently large and p is sufficiently small.

3.3.5 The uniform distribution

A continuous random variable X is said to follow a uniform distribution with parameters $a < b \in \mathbb{R}$, which is written as $X \sim Unif(a, b)$, if its probability density function and cumulative distribution function are defined as

$$f_X(x) = \frac{1}{b-a}\mathbb{I}_{[a,b]}(x) \qquad F_X(x) = \begin{cases} 0 & \text{if } x < a \\ \frac{x-a}{b-a} & \text{if } x \in [a, b] \\ 1 & \text{if } x > b \end{cases}$$

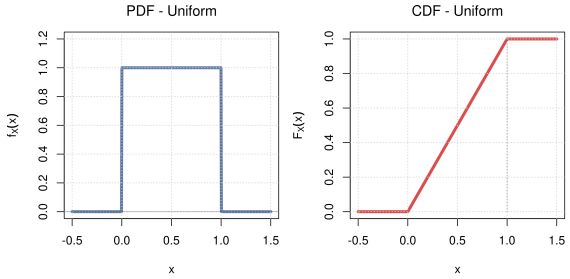


Figure 3.7: PDF and CDF for a random variable $X \sim Unif(0, 1)$.

An important application of the uniform distribution consists of generating random variables for simulation modelling.
Given a random variable $U \sim Unif(0, 1)$, let F be a cumulative distribution function with quantile function $\mathbb{Q}$: the "quantile transform" $Y = \mathbb{Q}(U)$ will be a random variable with cumulative distribution function $F_Y(y) = F$.

Proof:

Assuming F is continuous and strictly increasing, it is possible to exploit the fact that $\mathbb{Q} = F^{-1}$ to let

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(F^{-1}(U) \leq y) = \mathbb{P}(U \leq F(y)) = F(y), \text{ being } U \sim Unif(0, 1).$$

It should be noticed that this result is also valid whenever F is only weakly monotonic, although the proof is more cumbersome.

3.3.6 The exponential distribution

A continuous random variable X is said to follow an exponential distribution with parameter $\lambda \in \mathbb{R}^+$, which is written as $X \sim Exp(\lambda)$, if its probability density function and cumulative distribution

function are defined as

$$f_X(x) = \lambda e^{-\lambda x}\mathbb{I}_{\mathbb{R}^+}(x) \qquad F_X(x) = (1 - e^{-\lambda x})\mathbb{I}_{[0,+\infty)}(x)$$

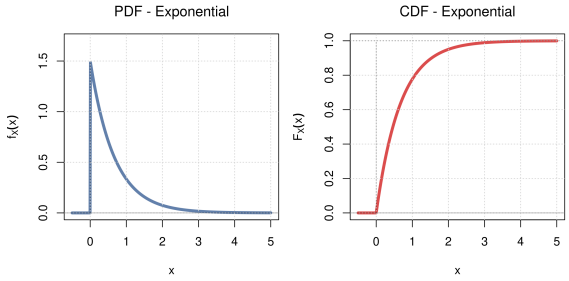


Figure 3.8: PDF and CDF for a random variable $X \sim Exp(1.5)$.

This distribution is often used to model survival and waiting times because, due to its memoryless behaviour, the elapsed time does not influence future outcomes, which means that

$$\mathbb{P}(X > x + k | X > x) = \mathbb{P}(X > k) \ \forall k > 0$$

Proof:

If $X \sim Exp(\lambda)$, then it is possible to observe that

$$\mathbb{P}(X > n) = 1 - \mathbb{P}(X \leq n) = 1 - F_X(n) = 1 - (1 - e^{-\lambda n}) = e^{-\lambda n} \ \forall n > 0$$

Therefore, by applying the definition of conditional probability, it is possible to conclude that

$$\mathbb{P}(X > x + k | X > x) = \frac{\mathbb{P}(\{X > x + k\} \cap \{X > x\})}{\mathbb{P}(X > x)} = \frac{e^{-\lambda(x+k)}}{e^{-\lambda x}} = e^{-\lambda k} = \mathbb{P}(X > k)$$

3.3.7 The normal distribution

A continuous random variable X is said to follow a normal distribution with parameters $\mu \in \mathbb{R}$ and $\sigma^2 \in \mathbb{R}^+$, which is written as $X \sim \mathcal{N}(\mu, \sigma^2)$, if its probability density function is defined as

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \text{ for } x \in \mathbb{R}.$$

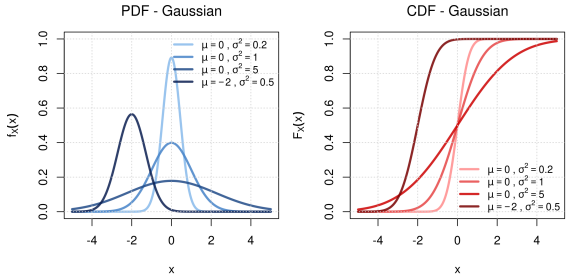


Figure 3.9: PDF and CDF for a random variable $X \sim \mathcal{N}(\mu, \sigma^2)$ for some choices of μ and σ^2.

This distribution has a symmetric, bell-shaped curve that is defined using the parameters μ, which represents the center and mode of the distribution, and σ, which describes its spread.
If $Z \sim \mathcal{N}(0,1)$, then Z is said to follow a "standard normal distribution" and its probability density function and cumulative distribution function are denoted by $\phi(z)$ and $\Phi(z)$, respectively.
In particular, one can transform a generic normal distribution into a standard normal distribution, and viceversa, through the following transformation:

$$X \sim \mathcal{N}(\mu, \sigma^2) \Rightarrow Z = \frac{X-\mu}{\sigma} \sim \mathcal{N}(0,1) \text{ and } Z \sim \mathcal{N}(0,1) \Rightarrow X = \sigma Z + \mu \sim \mathcal{N}(\mu, \sigma^2)$$

This transformation is useful because, starting from the standard normal table, it is possible to compute probabilities for a generic normal distribution as follows:

$$\begin{aligned}\mathbb{P}(X \in [a,b]) = \mathbb{P}(a \leq X \leq b) &= \mathbb{P}\left(\frac{a-\mu}{\sigma} \leq \frac{X-\mu}{\sigma} \leq \frac{b-\mu}{\sigma}\right) \\ &= \mathbb{P}\left(\frac{a-\mu}{\sigma} \leq Z \leq \frac{b-\mu}{\sigma}\right) = \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right)\end{aligned}$$

Example: Computing probabilities from a (standard) normal distribution

Suppose that, for a random variable $X \sim \mathcal{N}(3,5)$, the event $A = \{X > 1\}$ is observed: using standardization, it is possible to state that

$$\begin{aligned}\mathbb{P}(A) = \mathbb{P}(X > 1) = 1 - \mathbb{P}(X \leq 1) &= 1 - \mathbb{P}\left(Z \leq \frac{1-3}{\sqrt{5}}\right) \\ &= 1 - \mathbb{P}(Z \leq -0.8944) = 1 - \Phi(-0.8944) \approx 0.8144\end{aligned}$$

Similarly, if q_p and z_p denote the p-level quantiles of a generic and standard normal distribution, respectively, then it is possible to exploit the following relationship:

$$q_p = \sigma z_p + \mu, \text{ assuming that } X \sim \mathcal{N}(\mu, \sigma^2).$$

Overall, one of the biggest limitations of the normal distribution lies in the fact that, due to its lighter tails, it tends to be very sensitive to extreme values.

3.3.8 Student's t distribution

A continuous random variable X is said to follow a t distribution with $\nu > 0$ degrees of freedom, which is written as $X \sim t(\nu)$, if its probability density function is defined as

$$f_X(x) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\nu\pi}\Gamma(\frac{\nu}{2})}\left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}, \text{ for } x \in \mathbb{R}.$$

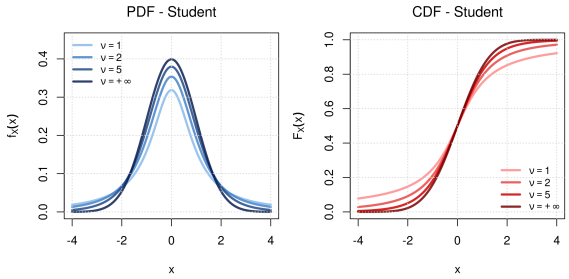


Figure 3.10: PDF and CDF for a random variable $X \sim t(\nu)$ for some choices of ν.

This distribution is often used as a more robust alternative to the normal distribution because its degrees of freedom allow to define heavier tails, although the t distribution converges to a standard normal distribution as $\nu \to +\infty$.

3.3.9 The Laplace distribution

A continuous random variable X is said to follow a Laplace distribution with parameters $\mu \in \mathbb{R}$ and $\lambda \in \mathbb{R}^+$, which is written as $X \sim Laplace(\mu, \lambda)$, if its probability density function and cumulative distribution function are defined as

$$f_X(x) = \frac{1}{2\lambda}e^{-\frac{|x-\mu|}{\lambda}} \qquad F_X(x) = \begin{cases} \frac{1}{2}e^{\frac{x-\mu}{\lambda}} & \text{if } x < \mu \\ 1 - \frac{1}{2}e^{-\frac{x-\mu}{\lambda}} & \text{if } x \geq \mu \end{cases}$$

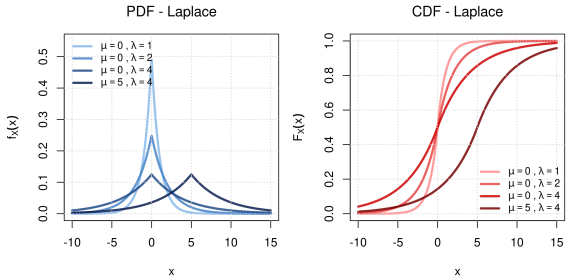


Figure 3.11: PDF and CDF for a random variable $X \sim Laplace(\mu, \lambda)$ for some choices of μ and λ.

This distribution is often used to introduced sparsity in statistical models thanks to its heavier tails.

3.3.10 The Gamma distribution

A continuous random variable X is said to follow a Gamma distribution with parameters $\alpha, \lambda > 0$, which is written as $X \sim Gamma(\alpha, \lambda)$, if its probability density function and cumulative distribution function are defined as

$$f_X(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} \mathbb{I}_{\mathbb{R}^+}(x) \qquad F_X(x) = \frac{\gamma(\alpha, \lambda x)}{\Gamma(\alpha)}$$

$$\Gamma(\alpha) = \int_0^{+\infty} t^{\alpha-1} e^{-t}\, dt \qquad \gamma(\alpha, x) = \int_0^x t^{\alpha-1} e^{-t}\, dt$$

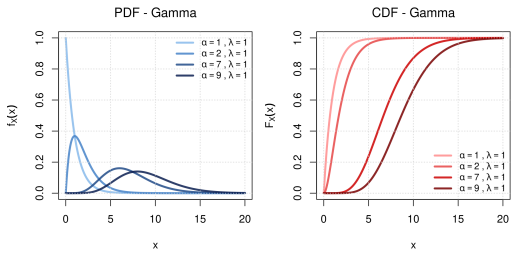


Figure 3.12: PDF and CDF for a random variable $X \sim Gamma(\alpha, \lambda)$ for some choices of α and λ.

For certain choices of α and λ, this distribution can be traced back to other known distributions, such as the exponential distribution or the chi-squared distribution.

3.3.11 The Beta distribution

A continuous random variable X is said to follow a Beta distribution with parameters $\alpha, \beta > 0$, which is written as $X \sim Beta(\alpha, \beta)$, if its probability density function and cumulative distribution function are defined as

$$f_X(x) = \frac{1}{Beta(\alpha, \beta)} x^{\alpha-1}(1-x)^{\beta-1} \mathbb{I}_{[0,1]}(x) \qquad F_X(x) = \frac{Beta(x; \alpha, \beta)}{Beta(\alpha, \beta)}$$

$$Beta(\alpha, \beta) = \int_0^1 t^{\alpha-1}(1-t)^{\beta-1}\, dt = \frac{(\alpha-1)!(\beta-1)!}{(\alpha+\beta-1)!} \qquad Beta(x; \alpha, \beta) = \int_0^x t^{\alpha-1}(1-t)^{\beta-1}\, dt$$

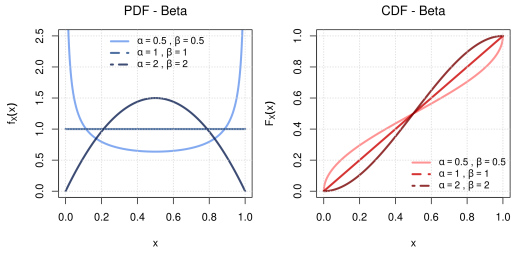


Figure 3.13: PDF and CDF for a random variable $X \sim Beta(\alpha, \beta)$ for some choices of α and β.

Chapter 4

Random vectors

4.1 Defining random vectors

Given a probability space $(\Omega, \mathbb{P})$, a random vector, also known as a p-variate random variable, is a (measurable) mapping $\mathbf{X} : \Omega \to \mathbb{R}^p$ that maps every outcome $\omega \in \Omega$ onto a vector $\mathbf{X}(\omega) \in \mathbb{R}^p$.
Generally speaking, the behaviour of a random vector, whose components can be studied individually through univariate analysis or jointly by analysing dependencies between single variables, is defined by its joint distribution, which is normally represented as a contingency table or tensor.
Furthermore, the joint distribution of a random vector can be used to derive the marginal distributions, which describe the behaviour of a single variable, and to model conditional distributions, which describe the behaviour of a variable conditioned to a given event.
Overall, random vectors can be seen as a generalization of random variables and, while they are mainly used to study bivariate distributions, their behaviour and properties can be extended to analyse $p > 2$ variables as well.

4.2 Fully discrete random vectors

A random vector is said to be "fully discrete" if all its components are discrete random variables.
For a bivariate distribution, the behaviour of a random vector $\mathbf{Z}$ is described using a joint probability mass function

$$p_{\mathbf{Z}}(\mathbf{z}) = p_{X,Y}(x, y) = \mathbb{P}(\{X = x\} \cap \{Y = y\}) \ \forall (x, y) \in S_X \times S_Y$$

Starting from the joint distribution, it is possible to retreive the marginal distribution of a single variable by computing

$$p_X(x) = \mathbb{P}\left(\{X = x\} \cap \left(\bigcup_{y \in S_Y} \{Y = y\}\right)\right) = \bigcup_{y \in S_Y} \mathbb{P}(\{X = x\} \cap \{Y = y\}) = \sum_{y \in S_Y} p_{X,Y}(x, y)$$

$$p_Y(y) = \mathbb{P}\left(\left(\bigcup_{x \in S_X} \{X = x\}\right) \cap \{Y = y\}\right) = \bigcup_{x \in S_X} \mathbb{P}(\{X = x\} \cap \{Y = y\}) = \sum_{x \in S_X} p_{X,Y}(x, y)$$

Furthermore, assuming a certain event has occurred, it is possible to model the conditional distribution of each variable as

$$p_{X|Y}(x|y) = \mathbb{P}(X = x|Y = y) = \frac{\mathbb{P}(\{X = x\} \cap \{Y = y\})}{\mathbb{P}(Y = y)} = \frac{p_{X,Y}(x, y)}{p_Y(y)} \ \forall x \in S_X$$

$$p_{Y|X}(y|x) = \mathbb{P}(Y = y|X = x) = \frac{\mathbb{P}(\{X = x\} \cap \{Y = y\})}{\mathbb{P}(X = x)} = \frac{p_{X,Y}(x, y)}{p_X(x)} \ \forall y \in S_Y$$

Example: The relationship between exam results and class attendance

Consider the random variables $X = \{\text{I pass the exam}\}$, whose support set is $S_X = \{0, 1\}$, and $Y = \{\text{I attend classes}\}$, whose support set is $S_Y = \{N, S, A\}$: on the set $S_X \times S_Y$, their joint distribution is defined as

	Y			
X	N	S	A	**Total**
0	0.6	0.05	0.05	0.7
1	0.2	0.01	0.09	0.3
Total	0.8	0.06	0.14	1.0

Looking at the contingency table, it is possible to recover the joint distributions

$$p_X(x) = \sum_{y \in S_Y} p_{X,Y}(x, y) = \begin{cases} 0.7 & \text{if } x = 0 \\ 0.3 & \text{if } x = 1 \end{cases}$$

$$p_Y(y) = \sum_{x \in S_X} p_{X,Y}(x, y) = \begin{cases} 0.8 & \text{if } y = N \\ 0.06 & \text{if } y = S \\ 0.14 & \text{if } y = A \end{cases}$$

Furthermore, it is possible to define the conditional distribution of X given the event $\{Y = y\}$, with $y \in S_Y$, as

$$p_{X|Y}(x|Y = N) = \mathbb{P}(X = x|Y = N) = \begin{cases} 0.75 & \text{if } x = 0 \\ 0.25 & \text{if } x = 1 \end{cases}$$

$$p_{X|Y}(x|Y = S) = \mathbb{P}(X = x|Y = S) = \begin{cases} 0.83 & \text{if } x = 0 \\ 0.17 & \text{if } x = 1 \end{cases}$$

$$p_{X|Y}(x|Y = A) = \mathbb{P}(X = x|Y = A) = \begin{cases} 0.36 & \text{if } x = 0 \\ 0.64 & \text{if } x = 1 \end{cases}$$

4.2.1 Defining independence for discrete random variables

When working with discrete random vectors, two random variables X and Y are independent if and only if their joint probability mass function factorises to the product of their marginal probability mass functions, meaning that

$$p_{X,Y}(x, y) = p_X(x)p_Y(y) \ \forall (x, y) \in S_X \times S_Y$$

Alternatively, it is possible to state that two random variables X and Y are independent if and only if their conditional distributions correspond to the marginal distributions, meaning that

$$p_{X|Y}(x|y) = p_X(x) \qquad p_{Y|X}(y|x) = p_Y(y)$$

Example: Exam results and class attendance as independent random variables

Consider the random variables $X = \{\text{I pass the exam}\}$, whose support set is $S_X = \{0, 1\}$, and $Y = \{\text{I attend classes}\}$, whose support set is $S_Y = \{N, S, A\}$: assuming the two random variables are independent, their joint distribution becomes

	Y			
X	N	S	A	**Total**
0	0.56	0.042	0.098	0.7
1	0.24	0.018	0.042	0.3
Total	0.8	0.06	0.14	1.0

Generally speaking, it is possible to check how much two random variables are far from independence by evaluating the distance between their joint distribution $p_{X,Y}$ and the factorised distribution $\tilde{p}$.

4.3 Fully continuous random vectors

A random vector is said to be "fully continuous" if all its components are continuous random variables.
For a bivariate distribution, the behaviour of a random vector $\mathbf{Z}$ is described using a joint probability density function such that

$$\mathbb{P}((X, Y) \in A) = \int \int_A f_{X,Y}(x, y)\ dxdy\ \forall A \subseteq S_X \times S_Y$$

In practice, a joint probability density function is typically constructed by normalizing a non-negative "kernel" function $g(x, y)$, resulting in

$$f_{X,Y}(x, y) = \frac{g(x, y)}{c}, \text{ with } c = \int \int_{\mathbb{R}^2} g(x, y)\ dxdy < +\infty$$

Starting from the joint distribution, it is possible to retrieve the marginal distribution of a single variable by computing

$$f_X(x) = \int_{S_Y} f_{X,Y}(x, y)\ dy$$
$$f_Y(y) = \int_{S_X} f_{X,Y}(x, y)\ dx$$

Furthermore, assuming a certain event has occurred, it is possible to model the conditional distribution of each variable as

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} \ \forall x \in S_X$$
$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)} \ \forall y \in S_Y$$

Alternatively, one can study the behaviour of a fully continuous random vector through the relationship between the joint cumulative distribution function and the joint probability density function:

$$F_{X,Y}(x,y) = \mathbb{P}(\{X \leq x\} \cap \{Y \leq y\}) = \int_{-\infty}^{y} \int_{-\infty}^{x} f_{X,Y}(u,v) \ dudv$$
$$f_{X,Y}(x,y) = \frac{\partial^2 F_{X,Y}(x,y)}{\partial x \partial y} = \frac{\partial^2 F_{X,Y}(x,y)}{\partial y \partial x}$$

Example: Deriving information from a fully continuous random vector

For two parameters $\alpha, \beta > 0$, consider the following joint cumulative distribution function:

$$F_{X,Y}(x,y) = 1 - e^{-\alpha x} - e^{-\beta y} + e^{-(\alpha x + \beta y)}, \text{ with } (x,y) \in \mathbb{R}^+ \times \mathbb{R}^+.$$

This means that the joint probability density function will be given by

$$f_{X,Y}(x,y) = \frac{\partial^2 F_{X,Y}(x,y)}{\partial x \partial y} = \frac{\partial[\alpha e^{-\alpha x} - \alpha e^{-(\alpha x + \beta y)}]}{\partial y} = \alpha\beta e^{-(\alpha x + \beta y)} \mathbb{I}_{\mathbb{R}^+ \times \mathbb{R}^+}(x,y)$$

Starting from the joint probability density function, it is possible to recover the following marginal probability density functions:

$$f_X(x) = \int_0^{+\infty} \alpha\beta e^{-(\alpha x + \beta y)} \ dy \overset{u=-\beta y}{=} \int_{-\infty}^{0} \alpha e^{-\alpha x}(-e^u) \ du = \alpha e^{-\alpha x} \mathbb{I}_{\mathbb{R}^+}(x)$$
$$f_Y(y) = \int_0^{+\infty} \alpha\beta e^{-(\alpha x + \beta y)} \ dy \overset{u=-\alpha x}{=} \int_{-\infty}^{0} \beta e^{-\beta y}(-e^u) \ du = \beta e^{-\beta y} \mathbb{I}_{\mathbb{R}^+}(y)$$

This means that $X \sim Exp(\alpha)$ and $Y \sim Exp(\beta)$.

4.3.1 Defining independence for continuous random variables

When working with continuous random vectors, two random variables X and Y are independent if and only if their joint probability density function factorises to the product of their marginal probability density functions, meaning that

$$f_{X,Y}(x,y) = f_X(x) f_Y(y) \ \forall (x,y) \in S_X \times S_Y$$

Alternatively, it is possible to state that two random variables X and Y are independent if and only if their conditional distributions correspond to the marginal distributions, meaning that

$$f_{X|Y}(x|y) = f_X(x) \qquad f_{Y|X}(y|x) = f_Y(y)$$

Example: The joint distribution of two independent exponential random variables

For two parameters $\alpha, \beta > 0$, consider the following joint cumulative distribution function:

$$F_{X,Y}(x,y) = 1 - e^{-\alpha x} - e^{-\beta y} + e^{-(\alpha x + \beta y)}, \text{ with } (x,y) \in \mathbb{R}^+ \times \mathbb{R}^+.$$

Knowing that $f_X(x) = \alpha e^{-\alpha x}\mathbb{I}_{\mathbb{R}^+}(x)$ and $f_Y(y) = \beta e^{-\beta y}\mathbb{I}_{\mathbb{R}^+}(y)$, observe that

$$\begin{aligned} f_{X,Y}(x,y) &= \frac{\partial^2 F_{X,Y}(x,y)}{\partial x \partial y} = \frac{\partial[\alpha e^{-\alpha x} - \alpha e^{-(\alpha x+\beta y)}]}{\partial y} = \alpha\beta e^{-(\alpha x+\beta y)}\mathbb{I}_{\mathbb{R}^+\times\mathbb{R}^+}(x,y) \\ &= \alpha e^{-\alpha x}\mathbb{I}_{\mathbb{R}^+}(x)\beta e^{-\beta y}\mathbb{I}_{\mathbb{R}^+}(y) = f_X(x)f_Y(y)\ \forall (x,y) \in \mathbb{R}^+ \times \mathbb{R}^+ \Rightarrow X \perp Y \end{aligned}$$

4.4 Mixed random vectors

A random vector is said to be "mixed" if some of its components are discrete random variables, whereas the other components are continuous random variables.
For a bivariate distribution, the behaviour of a random vector $\mathbf{Z}$ is described using a mixture model that can be defined as

$$f_{X,Y}(x,y) = f_{X|Y}(x|y)f_Y(y) = f_{Y|X}(y|x)f_X(x)\ \forall (x,y) \in S_X \times S_Y$$

In particular, assuming X is the discrete component while Y is the continuous component, it is possible to retrieve the marginal distribution of a single variable by computing

$$\begin{aligned} f_X(x) = p_X(x) &= \int_{S_Y} f_{X,Y}(x,y)\ dy = \int_{S_Y} f_{X|Y}(x|y)f_Y(y)\ dy \\ f_Y(y) &= \sum_{x \in S_X} f_{X,Y}(x,y) = \sum_{x \in S_X} f_{Y|X}(y|x)f_X(x) \end{aligned}$$

Example: Investigating adverse reactions in drugs

A team studies the behaviour of a new drug using a discrete random variable X, which counts the number of patients showing an adverse reaction to the drug, and a continuous random variable P, which indicates the probability of an adverse reaction to the drug.
On a sample of n patients, it is known that $(X|P=p) \sim Binom(n,p)$ and $P \sim Unif(0,1)$, allowing to marginalize

$$\begin{aligned} f_X(x) &= \int_{S_P} f(x|p)f(p)\ dp = \int_0^1 \binom{n}{x} p^x (1-p)^{n-x}\ dp \\ &\overset{\text{Beta integral}}{=} \frac{n!}{x!(n-x)!}\frac{x!(n-x)!}{(n+1)!} = \frac{1}{n+1} \end{aligned}$$

This means that, on its support $S_X = \{0, \ldots, n\}$, X follows a discrete uniform distribution, which is a special case of the Beta-binomial distribution.

An important application of mixed random vectors consists of studying the distributions that involve an observable random variable, which typically follows a simple distribution, and a latent random variable that models the variability of the sampled data.

Example: Finite mixture models

A bivariate random vector contains an observable discrete random variable X and a latent discrete random variable H whose distribution on the set $S_H = \{1, \dots, k\}$ is given by

$$\mathbb{P}(H = j) = \pi_j, \text{ with } \pi_j \geq 0 \ \forall j \in S_H \text{ and } \sum_{j=1}^{k} \pi_j = 1$$

This means that, if $(X|H = j) \sim f_j(x)$, the marginal distribution of X can be formulated as

$$f_X(x) = \sum_{i=1}^{k} f_j(x)\pi_j, \text{ which represents a finite mixture with } k \text{ sub-populations.}$$

Assuming that, on the support $S_X = \{0, 1\}$, $(X|H = j) \sim Ber(\pi_j)$, the marginal distribution of X can be modelled using the law of total probability:

$$\mathbb{P}(E) = \sum_{j=1}^{k} \mathbb{P}(E \cap H_j) = \sum_{j=1}^{k} \mathbb{P}(E|H_j)\mathbb{P}(H_j)$$

In fact, if E represents the observed value of X and H is used to model a sequence of mutually exclusive hypotheses, it is possible to recover that $X \sim Ber(\mathbb{P}(E))$.

4.5 The joint cumulative distribution function

The joint cumulative distribution function of any bivariate random vector can be defined as

$$F_{X,Y}(x, y) = \mathbb{P}(\{X \leq x\} \cap \{Y \leq y\}) \ \forall (x, y) \in S_X \times S_Y$$

In this case, it is possible to retrieve the marginal cumulative distribution function function of a single variable by computing

$$F_X(x) = \lim_{y \to +\infty} F_{X,Y}(x, y) \ \forall x \in S_X$$
$$F_Y(y) = \lim_{x \to +\infty} F_{X,Y}(x, y) \ \forall y \in S_Y$$

Furthermore, whenever two random variables X and Y are independent, it is the case that

$$F_{X,Y}(x, y) = F_X(x)F_Y(y) \ \forall (x, y) \in \mathbb{R}^2$$

4.5.1 Building independent and identically distributed random samples

A random sample $\mathcal{D}_n = \{X_1, \dots, X_n\}$ is said to be "independent and identically distributed" if all the sampled random variables are independent and come from the same population model F_X,

allowing to define their joint distribution using the likelihood function, resulting in

$$\mathbb{P}(\mathcal{D}_n) = F(x_1, \ldots, x_n) = \mathbb{P}(X_1 \leq x_1, \ldots, X_n \leq x_n) = \prod_{i=1}^{n} \mathbb{P}(X_i \leq x_i) = \prod_{i=1}^{n} F(x_i)$$

Example: Maximum likelihood estimation on a Gaussian mixture model

Suppose that the data points of a random sample $\mathcal{D}_n$ are drawn independent and identically distributed from a K-mode "mixture of Gaussians" distribution, meaning that

$$\mathbb{P}(x_i) = \sum_{j=1}^{K} \pi_j \cdot \mathcal{N}(x_i|, \mu_j, \sigma_j^2) \; \forall x_i \in \mathcal{D}$$

Under the assumption of independence, the joint probability of the random sample can be expressed using the likelihood function, resulting in

$$\mathbb{P}(\mathcal{D}_n) = \mathbb{P}(x_1, \ldots, x_n) = \prod_{i=1}^{n} \mathbb{P}(x_i) = \prod_{i=1}^{n} \left(\sum_{j=1}^{K} \pi_j \cdot \mathcal{N}(x_i|\mu_j, \sigma_j^2) \right)$$

If the parameters $\boldsymbol{\Theta} = \{(\pi_j, \mu_j, \sigma_j^2)\}_{j=1}^{K}$ are unknown, one can learn the optimal parameters by performing maximum likelihood estimation:

$$\hat{\boldsymbol{\Theta}} = \arg\max_{\boldsymbol{\Theta}} L(\boldsymbol{\Theta}) = \arg\max_{\boldsymbol{\Theta}} \mathbb{P}(\mathcal{D}_n)$$

Overall, a machine learning model would need to learn $(3K - 1)$ parameters, although the number of modes is often counted as an additional parameter, but some solutions can cause the model to reach infinite likelihood.

In some cases, however, it can be more convenient to express the likelihood function using a probability density function and a penalty or constraint on the data.

Chapter 5

Expectation and variance

5.1 Summarizing a distribution

While the distribution of a random variable provides all the probabilistic information about its behaviour, it can be cumbersome to analyse.
For this reason, a probability distribution is often summarized by looking at its quantiles, which give an idea of the possible values taken by the random variable, or using other parameters, such as the expected value or the variance, that provide information about the central behaviour and spread of the distribution.

5.2 The expected value of a random variable

Intuitively, the expected value of a random variable can be seen as a weighted average of its possible values with respect to their probabilities.
In fact, the expected value of a random variable $X : \Omega \to S_X$ with cumulative distribution F_X is formally defined as

$$\mathbb{E}(X) \int_{\mathbb{R}} x \; dF_X(x) = \begin{cases} \sum_{x \in S_X} x p_X(x) & \text{if } X \text{ is discrete and has PMF } p_X(x) \\ \int_{S_X} x f_X(x) \; dx & \text{if } X \text{ is continuous and has PDF } f_X(x) \end{cases}$$

Example: Modelling the average failure time of a device

The failure time of a device is modelled using a random variable X whose probability density function is given by $f_X(x) = 2x^{-3}\mathbb{I}_{[1,+\infty)}(x)$, meaning that the expected time to failure will be given by

$$\begin{aligned} \mathbb{E}(X) = \int_{\mathbb{R}} x f_X(x) \; dx = \int_{\mathbb{R}} x \frac{2}{x^3} \mathbb{I}_{[1,+\infty)}(x) \; dx = \int_1^{+\infty} \frac{2}{x^2} \; dx \\ = \lim_{L \to +\infty} \int_1^{+\infty} \frac{2}{x^2} \; dx = \lim_{L \to +\infty} -\frac{2}{x} \Big|_1^L = 2 \end{aligned}$$

In particular, if a random variable follows a distribution that is symmetric around a central value x_0, then $\mathbb{E}(X) = x_0$, provided the mean is well-defined.

Example: Exploiting symmetry in the uniform distribution

The uniform distribution is a popular symmetric distribution that is centered around the midpoint of the interval defined by its support.
In fact, if $X \sim Unif(a, b)$, then its expected value will be given by

$$\mathbb{E}(X) = \int_{\mathbb{R}} x \, dF_X(x) = \int_a^b \frac{x}{b-a} \, dx = \frac{x^2}{2(b-a)}\bigg|_a^b = \frac{b^2-a^2}{2(b-a)} = \frac{(b-a)(b+a)}{2(b-a)} = \frac{a+b}{2}$$

Overall, the expected value of a random variable depends solely on its distribution and can be seen as a "center of mass", although it is not robust to extreme values because a small change in the probability of small or large values can significantly affect the expected value.

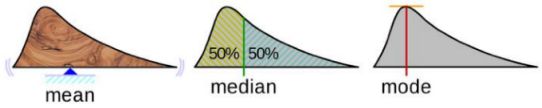


Figure 5.1: Comparing the mean, median and mode of a distribution.

Furthermore, whenever the expected value of a distribution does not exist because it is not well-defined, it can be more convenient to analyse the median instead.

Example: The Cauchy distribution

A continuous random variable X is said to follow a Cauchy distribution if $X \sim t(1)$, meaning that its probability density function will be given by

$$f_X(x) = \frac{1}{\pi(x^2+1)}, \text{ for } x \in \mathbb{R}.$$

Looking at its heavy-tailed shape, one can discover that

$$\begin{aligned}
\mathbb{E}(X) &= \int_{-\infty}^{+\infty} \frac{x}{\pi(x^2+1)} \, dx = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \frac{2x}{x^2+1} \, dx = \frac{1}{2\pi} \left(\int_{-\infty}^{0} \frac{2x}{x^2+1} \, dx + \int_0^{+\infty} \frac{2x}{x^2+1} \, dx \right) \\
&= \frac{1}{2\pi} \left[\left(\lim_{a \to -\infty} \int_a^0 \frac{2x}{x^2+1} \, dx \right) + \left(\lim_{b \to +\infty} \int_0^b \frac{2x}{2^2+1} \, dx \right) \right] \\
&= \frac{1}{2\pi} \left[\left(\lim_{a \to -\infty} \ln(x^2+1) \Big|_a^0 \right) + \left(\lim_{b \to +\infty} \ln(x^2+1) \Big|_0^b \right) \right] = -\infty + \infty
\end{aligned}$$

5.2.1 The law of the lazy statistician

The law of the lazy statistician states that, if X is a (continuous) random variable with cumulative distribution function F_X, then, for a deterministic function r, the expected value of the random variable $Y = r(X)$ is given by

$$\mathbb{E}(Y) = \int_{\mathbb{R}} y \; dF_y(y) = \mathbb{E}(r(X)) = \int_{\mathbb{R}} r(x) \; dF_X(x)$$

Example: Probabilities as expected values

Assuming X is a continuous random variable with probability density function $f_X(x)$, pick any event A and define the indicator variable $Y = \mathbb{I}_A(X)$: the expected value of Y will be given by

$$\mathbb{E}(Y) = \mathbb{E}(\mathbb{I}_A(X)) = \int_{\mathbb{R}} \mathbb{I}_A(x) f_X(x) \; dx = \int_A f_X(x) \; dx = \mathbb{P}(X \in A)$$

However, one has $\mathbb{E}(r(X)) = r(\mathbb{E}(X))$ if and only if r is an affine map.
In fact, if $Y = r(X) = \alpha + \beta X$, it is possible to state that

$$\mathbb{E}(Y) = \mathbb{E}(r(X)) = \mathbb{E}(\alpha + \beta X) = \alpha + \beta X = r(\mathbb{E}(X))$$

Proof:

If $Y = \alpha + \beta X$, where X is a random variable with cumulative distribution function $F_X(x)$, then, by the law of the lazy statistician, one has

$$\begin{aligned} \mathbb{E}(Y) = \int_{\mathbb{R}} (\alpha + \beta x) \; dF_X(x) &= \int_{\mathbb{R}} \alpha \; dF_X(x) + \int_{\mathbb{R}} \beta x \; dF_X(x) \\ &= \alpha \int_{\mathbb{R}} dF_X(x) + \beta \int_{\mathbb{R}} x \; dF_X(x) = \alpha + \beta \mathbb{E}(X) \end{aligned}$$

In general, the behaviour of a non-affine map depends on its convexity.
A function $r : \mathbb{R} \to \mathbb{R}$ is said to be convex on a set $S \subseteq \mathbb{R}$ if, for any $x_1, x_2 \in S$ and $t \in [0, 1]$, one has

$$r(tx_1 + (1-t)x_2) \leq tr(x_1) + (1-t)r(x_2)$$

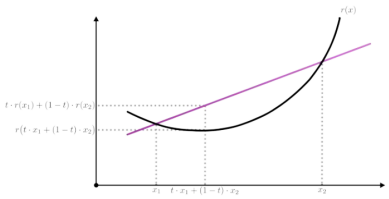


Figure 5.2: Visualization of a convex function.

In particular, Jensen's inequality states that, if r is a (strictly) convex deterministic function, it must be the case that $r(\mathbb{E}(X)) \leq \mathbb{E}(r(X))$.

Example: The evidence lower bound for a finite mixture of Gaussians

Assuming that, given a generic parametrization $\boldsymbol{\Theta}$, the joint distribution of a random sample $\mathcal{D}_n$ is modelled as a K-mode mixture of Gaussians, one can learn $\boldsymbol{\Theta}$ by analysing the likelihood function, meaning that

$$\hat{\boldsymbol{\Theta}} = \arg\max_{\boldsymbol{\Theta}} L(\boldsymbol{\Theta}), \text{ where } L(\boldsymbol{\Theta}) = \mathbb{P}_{\boldsymbol{\Theta}}(\mathcal{D}_n) = \prod_{i=1}^{n} p_{\boldsymbol{\Theta}}(x_i) = \prod_{i=1}^{n} \left(\sum_{j=1}^{K} \pi_j \cdot \mathcal{N}(x_i | \mu_j, \sigma_j^2) \right)$$

Optimizing this function can be made easier by introducing a latent random variable Y, which is assumed to follow a marginal distribution q, and formulate

$$\log(p_{\boldsymbol{\Theta}}(x)) = \log\left(\int_{S_Y} p_{\boldsymbol{\Theta}}(x,y)\ dy \right) = \log\left(\int q(y) \frac{p_{\boldsymbol{\Theta}}(x,y)}{q(y)}\ dy \right) = \log\left(\mathbb{E}_q\left(\frac{p_{\boldsymbol{\Theta}}(x,y)}{q(y)} \right) \right)$$

Since $f(x) = \log(x)$ is a concave function, Jensen's inequality implies that

$$\log(p_{\boldsymbol{\Theta}}(x)) = \log\left(\mathbb{E}_q\left(\frac{p_{\boldsymbol{\Theta}}(x,y)}{q(y)} \right) \right) \geq \mathbb{E}_q\left(\log\left(\frac{p_{\boldsymbol{\Theta}}(x,y)}{q(y)} \right) \right) = \mathcal{F}(q, \boldsymbol{\Theta})$$

Therefore, it is possible to define the information gap as

$$\begin{aligned}
\log(p_{\boldsymbol{\Theta}}(x)) - \mathcal{F}(q, \boldsymbol{\Theta}) &= \log(p_{\boldsymbol{\Theta}}(x)) - \int_{S_Y} q(y) \log\left(\frac{p_{\boldsymbol{\Theta}}(x,y)}{q(y)} \right) dy \\
&= \log(p_{\boldsymbol{\Theta}}(x)) - \int_{S_Y} q(y) \log\left(\frac{p_{\boldsymbol{\Theta}}(y|x) p_{\boldsymbol{\Theta}}(x)}{q(y)} \right) dy \\
&= \log(p_{\boldsymbol{\Theta}}(x)) - \left(\int_{S_Y} q(y) \log\left(\frac{p_{\boldsymbol{\Theta}}(y|x)}{q(y)} \right) dy + \int_{S_Y} q(y) \log(p_{\boldsymbol{\Theta}}(x))\ dy \right) \\
&= \log(p_{\boldsymbol{\Theta}}(x)) - \left(\int_{S_Y} q(y) \log\left(\frac{p_{\boldsymbol{\Theta}}(y|x)}{q(y)} \right) dy + \log(p_{\boldsymbol{\Theta}}(x)) \right) \\
&= -\int_{S_Y} q(y) \log\left(\frac{p_{\boldsymbol{\Theta}}(y|x)}{q(y)} \right) dy = \int_{S_Y} q(y) \log\left(\frac{q(y)}{p_{\boldsymbol{\Theta}}(y|x)} \right) dy
\end{aligned}$$

The computations reveal that the information gap can be expressed as a Kullback-Leiber divergence, which can be proved to be minimum for $q(y) = p_{\boldsymbol{\Theta}}(y|x)$ using the expectation maximization algorithm on the parametric model $\mathcal{F}(q, \boldsymbol{\Theta})$.

5.2.2 Variational characterizations for random variables

Suppose you want to predict a value d for a random variable X, which is assumed to have finite mean μ and median m.
Generally speaking, the quality of a prediction can be measured using a loss function $\mathcal{L}(X, d)$ that quantifies the resulting prediction error.
By considering the risk function, it is possible to identify the optimal prediction as the value d^* that minimizes the average prediction error, meaning that

$$d^* = \arg\min_d \mathbb{E}(\mathcal{L}(X, d))$$

For instance, whenever the mean squared error is used to evaluate a regression model, it is possible to show that $\mathbb{E}((X-\mu)^2) \leq \mathbb{E}((X-d)^2) \Rightarrow d^* = \mu$.

Proof:

Using the linearity of expectation, let

$$\mathbb{E}((X-d)^2) = \mathbb{E}(X^2 - 2dX + d^2) = \mathbb{E}(X^2) + \mathbb{E}(-2dX) + \mathbb{E}(d^2) = \mathbb{E}(X^2) - 2d\mu + d^2$$

Optimizing the mean squared error with respect to d results in

$$\frac{\partial \mathbb{E}((X-d)^2)}{\partial d} = -2\mu + 2d = 0 \Leftrightarrow d^* = \mu$$

On the other hand, whenever the model is evaluated using the mean absolute error, one has $\mathbb{E}(|X-m|) \leq \mathbb{E}(|X-d|) \Rightarrow d^* = m$.

Proof:

Let X be a continuous random variable with probability density function f_X: assuming $d > m$, consider

$$\begin{aligned}
\mathbb{E}(|X-d|) - \mathbb{E}(|X-m|) &= \int_{-\infty}^{+\infty} (|x-d| + |x-m|) f_X(x)\, dx \\
&\geq \int_{-\infty}^{m} (d-m) f_X(x) dx + \int_{m}^{d} (m-d) f_X(x) dx + \int_{d}^{+\infty} (m-d) f_X(x) dx \\
&= (d-m)(\mathbb{P}(X \leq m) - \mathbb{P}(X > m)) \overset{\mathbb{P}(X\leq m)\geq \frac{1}{2}\geq \mathbb{P}(X>m)}{\geq} 0
\end{aligned}$$

This result suggests that $\mathbb{E}(|X-m|) \leq \mathbb{E}(|X-d|)$, meaning that $d^* = m$ acts as a minimum for the mean absolute error.

Overall, variational characterizations come in handy to define an upper bound for the deviation between the mean and median of a (well-defined) random variable.

Proof:

For a well-defined random variable X, which has mean equal to $\mathbb{E}(X) = \mu$ and median m, it is possible to state that

$$|\mu - m| = |\mathbb{E}(X) - m| \overset{r(x)=|x|}{=} r(\mathbb{E}(X) - m) \overset{\text{Jensen}}{\leq} \mathbb{E}(r(X - m)) = \mathbb{E}(|X - m|)$$
$$\leq \mathbb{E}(|X - \mu|) \overset{\sqrt{x^2}=|x|}{=} \mathbb{E}\left(\sqrt{(X - \mu)^2}\right) \overset{\text{Jensen - Concave}}{\leq} \sqrt{\mathbb{E}((X - \mu)^2)} = \sigma$$

Therefore, the standard deviation of the random variable also acts as an upper bound for the fluctuation between its mean and median.

5.3 The variance of a random variable

Intuitively, the variance of a random variable can describe how a representative for its distribution, such as the expected value, tends to deviate from the typical behaviour.
In fact, if a random variable $X : \Omega \to S_X$ with cumulative distribution function F_X has finite mean $\mathbb{E}(X) = \mu$, then its variance is formally defined as

$$\mathbb{V}(X) = \mathbb{E}((X - \mathbb{E}(X))^2) = \int_{\mathbb{R}} (x - \mu)^2 \, dF_X(x) \Rightarrow SD(X) = \sqrt{\mathbb{V}(X)}$$

Example: Modelling price variability

Suppose that the prices of two services are modelled using the random variables $X \sim Unif(25, 35)$ and $Y \sim Unif(15, 45)$: one can see that

$$\mathbb{E}(X) = \frac{25 + 35}{2} = 30 = \frac{15 + 45}{2} = \mathbb{E}(Y)$$

However, looking at the variance of each variable reveals that

$$\mathbb{V}(X) = \int_{\mathbb{R}} (x - 30)^2 \frac{1}{10} \mathbb{I}_{[25,35]}(x) \, dx = \int_{25}^{35} \frac{(x - 30)^2}{10} \, dx = \frac{25}{3}$$
$$\mathbb{V}(Y) = \int_{\mathbb{R}} (x - 30)^2 \frac{1}{30} \mathbb{I}_{[15,45]}(x) \, dx = \int_{15}^{45} \frac{(x - 30)^2}{30} \, dx = 75$$

Alternatively, one can use the linearity property to express the variance of a random variable as

$$\mathbb{V}(X) = \mathbb{E}((X - \mathbb{E}(X))^2) = \mathbb{E}(X^2 - 2X\mathbb{E}(X) + \mathbb{E}^2(X))$$
$$= \mathbb{E}(X^2) - 2\mathbb{E}(X)\mathbb{E}(X) + \mathbb{E}^2(X) = \mathbb{E}(X^2) - \mathbb{E}^2(X)$$

In particular, if $Y = \alpha + \beta X$, then it is possible to state that $\mathbb{V}(Y) = \beta^2 \mathbb{V}(X)$.

Proof:

If $Y = \alpha + \beta X$, where X has (finite) mean $\mathbb{E}(X)$, then its variance is given by

$$\begin{aligned}\mathbb{V}(Y) = \mathbb{V}(\alpha + \beta X) &= \mathbb{E}((\alpha + \beta X - \mathbb{E}(\alpha + \beta X))^2) = \mathbb{E}((\alpha + \beta X - \alpha - \beta\mathbb{E}(X))^2) \\ &= \mathbb{E}((\beta(X - \mathbb{E}(X)))^2) = \mathbb{E}(\beta^2(X - \mathbb{E}(X))^2) = \beta^2\mathbb{V}(X)\end{aligned}$$

Furthermore, if X is a bounded random variable, then its variance must exist and be finite, with $\mathbb{V}(X) = 0$ if and only if $X = \mathbb{E}(X)$.

Proof:

If $X = \mathbb{E}(X)$, then, trivially, one has

$$\mathbb{V}(X) = \mathbb{E}((X - \mathbb{E}(X))^2) = \mathbb{E}((\mathbb{E}(X) - \mathbb{E}(X))^2) = \mathbb{E}(0) = 0$$

Conversely, if $\mathbb{V}(X) = 0$, one can fix $a > 0$ and consider

$$\begin{aligned}0 = \mathbb{V}(X) = \mathbb{E}((X - \mathbb{E}(X))^2) &\geq \mathbb{E}((X - \mathbb{E}(X))^2 \mathbb{I}_{|X - \mathbb{E}(X)| \geq a}(x)) \\ &\geq a^2(\mathbb{E}(\mathbb{I}_{|X - \mathbb{E}(X)| \geq a}(x)) \\ &= a^2\mathbb{P}(|X - \mathbb{E}(X)| \geq a) \overset{a>0}{\Rightarrow} \mathbb{P}(|X - \mathbb{E}(X)| \geq a) = 0\end{aligned}$$

Knowing that $\mathbb{V}(X) = 0$, it must be the case that

$$\begin{aligned}\mathbb{P}(X = \mathbb{E}(X)) = \mathbb{P}(X - \mathbb{E}(X) = 0) &= 1 - \mathbb{P}(|X - \mathbb{E}(X)| > 0) \\ &= \lim_{a \to 0^+} 1 - \mathbb{P}(|X - \mathbb{E}(X)| \geq a) = 1 \Rightarrow X = \mathbb{E}(X)\end{aligned}$$

Overall, the variance of a random variable depends solely on its distribution, implying that it may not exist under some circumstances in which it is not well-defined.
For this reason, the mean absolute deviation can provide a more robust measure for spread.

5.4 The moment of a random variable

The kth moment of a random variable X with cumulative distribution function F_X is defined as the quantity $\mu_k = \mathbb{E}(X^k)$, provided $\mathbb{E}(|X|^k) < +\infty$.

Example: Moments as distribution summaries

If well-defined, the first moment of any random variable X corresponds to its expected value. Additionally, the third moment of a random variable allows to describe the asymmetry of its distributions, while the fourth moment comes in handy to measure the heaviness of the tails of the distribution.

In particular, if the kth moment of a random variable X exists, then, for any $j < k$, the jth moment exists as well.

Proof:

Assuming X has cumulative distribution function F_X, define

$$\begin{aligned}\mathbb{E}(|X|^j) = \int_{\mathbb{R}} |x|^j \; dF_X(x) &= \int_{|x|\leq 1} |x|^j \; dF_X(x) + \int_{|x|>1} |x|^j \; dF_X(x) \\ &\leq \int_{|x|\leq 1} dF_X(x) + \int_{|x|>1} |x|^k \; dF_X(x) \leq 1 + \mathbb{E}(|X|^k) < +\infty\end{aligned}$$

5.5 The (marginal) expected value of a random vector

The expected value of a random vector $\mathbf{X} \in \mathbb{R}^p$ with joint cumulative distribution function $F_{\mathbf{X}}$ can be trivially expressed by the coordinate-wise mean vector, meaning that

$$\mu = \mathbb{E}(\mathbf{X}) = \begin{pmatrix} \mathbb{E}(X_1) \\ \vdots \\ \mathbb{E}(X_p) \end{pmatrix} = \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_p \end{pmatrix}$$

In particular, the expected value of a (continuous) multivariate distribution is linear.
In fact, given a linear predictor $Y = \alpha^T \mathbf{X}$, the law of the lazy statistician allows to state that

$$\begin{aligned}\mathbb{E}(Y) = \mathbb{E}(r(X_1, \ldots, X_p)) &= \int \cdots \int r(x_1, \ldots, x_p) \; dF_{\mathbf{X}}(x_1, \ldots, x_p) \\ &= \int \cdots \int \alpha^T \mathbf{x} f_{\mathbf{X}}(\mathbf{x}) \; d\mathbf{x} \\ &= \sum_{j=1}^{p} \int_{\mathbb{R}} \alpha_j x_j f_{X_j}(x_j) \; dx_j = \sum_{j=1}^{p} \alpha_j \mathbb{E}(X_j) = \alpha^T \mu\end{aligned}$$

Furthermore, if all the components of a random vector are independent, then it is the case that

$$\mathbb{E}(Y) = \mathbb{E}\left(\prod_{j=1}^{p} X_i \right) = \int \cdots \int \prod_{j=1}^{p} x_j f_{\mathbf{X}}(\mathbf{x}) \; d\mathbf{x} = \prod_{j=1}^{p} \int_{\mathbb{R}} x_j f_{X_j}(x_j) \; dx_j = \prod_{j=1}^{p} \mathbb{E}(X_j)$$

5.5.1 The law of total expectation

For a bivariate distribution involving two random variables X and Y, the conditional expectations given fixed values of the conditioning variable are defined as

$$\begin{aligned}\mathbb{E}(Y|X = x) &= \int y \; dF_{Y|X}(y|x) \\ \mathbb{E}(X|Y = y) &= \int x \; dF_{X|Y}(x|y)\end{aligned}$$

Whenever the conditioning variable acts as a random object, the law of total expectation allows to recover the marginal expectations by considering

$$\mathbb{E}_Y(Y) = \mathbb{E}_X(\mathbb{E}_{Y|X}(Y|X))$$
$$\mathbb{E}_X(X) = \mathbb{E}_Y(\mathbb{E}_{X|Y}(X|Y))$$

Proof:

Assuming X and Y are continuous random variables, let

$$\begin{aligned}\mathbb{E}_X(\mathbb{E}_{Y|X}(Y|X)) &= \int r(x) f_X(x)\, dx = \int \left(\int y f_{Y|X}(y|x)\, dy \right) f_X(x)\, dx \\ &= \int \left(\int y \frac{f_{X,Y}(x,y)}{f_X(x)}\, dy \right) f_X(x)\, dx = \int \left(\int y f_{X,Y}(x,y)\, dx \right) dy \\ &\overset{\text{Marginalization}}{=} \int y f_Y(y)\, dy = \mathbb{E}_Y(Y)\end{aligned}$$

Note that, without loss of generality, the same strategy can be used for $\mathbb{E}_X(X)$.

5.6 The (marginal) variance of a random vector

The variance of a random vector often comes in handy to model the variability of its components or to study how they interact.
In fact, given a random vector $\mathbf{X} \in \mathbb{R}^p$ with component-wise mean μ, if $Y = \alpha^T \mathbf{X}$, then one has

$$\begin{aligned}\mathbb{V}(Y) = \mathbb{V}(\alpha^T \mathbf{X}) &= \mathbb{E}((\alpha^T \mathbf{X})^2) - \mathbb{E}^2(\alpha^T \mathbf{X}) \\ &= \mathbb{E}((\alpha^T \mathbf{X})^2) - (\alpha^T \mathbf{X})^2 = \mathbb{E}(\alpha^T \mathbf{X}\mathbf{X}^T \alpha) - \alpha^T \mu\mu^T \alpha \\ &= \alpha^T (\mathbb{E}(\mathbf{X}\mathbf{X}^T))\alpha - \alpha^T (\mu\mu^T)\alpha = \alpha^T (\mathbb{E}(\mathbf{X}\mathbf{X}^T) - \mu\mu^T)\alpha \\ &= \alpha^T Cov(\mathbf{X})\alpha, \text{ where } Cov(\mathbf{X}) \text{ is a } (p \times p) \text{ symmetric "covariance matrix".}\end{aligned}$$

Knowing that $(\mu\mu^T)_{ij} = \mu_i \mu_j$ and $(\mathbb{E}(\mathbf{X}\mathbf{X}^T))_{ij} = \mathbb{E}(X_i X_j)$, a generic entry of the covariance matrix will be given by

$$Cov(X_i, X_j) = \mathbb{E}(X_i X_j) - \mu_i \mu_j = \mathbb{E}((X_i - \mu_i)(X_j - \mu_j))$$

Therefore, the variance of a many-to-one linear predictor can be expressed as

$$\mathbb{V}(Y) = \mathbb{V}(\alpha^T \mathbf{X}) = \alpha^T Cov(\mathbf{X})\alpha = \sum_{j=1}^{p} \alpha_j^2 \mathbb{V}(X_j) + 2 \sum_{i<j} \alpha_i \alpha_j Cov(X_i, X_j)$$

Chapter 6

Stochastic simulations

6.1 Defining simulations

A stochastic simulation allows to study complex probabilistic models by generating values for a random variable that will be used into the model, providing an approximation of a random system. Simulations come in handy to investigate the typical behaviour of a system and its fluctuations, although more advanced techniques can be integrated to study exceptional cases as well.

Example: A physical simulation

Suppose you want to create a simulation of a balanced coin flip based on the result seen by tossing a fair die: a possible strategy could consist of having the coin show heads when the die shows an odd result, or tails when the die shows an even result.
Alternatively, it is possible to simulate the result shown by the die by (repeatedly) tossing the coin and associating each sequence of outcomes to a certain value.

6.2 Pseudorandom number generators

A pseudorandom number generator is a deterministic algorithm that, starting from an initial "seed" u_0, produces a sequence of numbers that resembles a random sequence.
For this reason, a pseudorandom number generator can be modelled as a function $\mathbf{f} : (0,1) \to (0,1)$ such that, for any seed u_0, the resulting sequence

$\{u_0, \mathbf{f}(u_0), \dots, \mathbf{f}^n(u_0)\}$ behaves like a $Unif(0,1)$ independent and identically distributed sequence.

A common method for pseudorandom number generation is the linear congruential generator, which, on the set $\{1, \dots, M\}$, is defined as

$$\mathbf{f}(x) = (\alpha + \beta x) \bmod M$$

```
lcg <- function(u0, n=1000, a=69069069, b=12345, M=2^32){
  x_out <- rep(NA, n)
```

```
    x_out[1] <- u0
    for (i in 2:n) {
      x_out[i] <- (a + (b * x_out[i - 1])) %% M
    }
    return(x_out / (M + 1))
  }
```

Listing 6.1: R implementation of a linear congruential generator.

In fact, this function acts as a generator on $(0, 1)$ when scaled by $(M + 1)$.

Example: A simple linear congruential generator

Consider the linear congruential generator

$$\mathbf{f}(x) = (69069069x + 12345) \bmod 2^{32}$$

For a given u_0, the function generates the sequence

$$518974515, 2498053016, 1113825472, 1109377984, \ldots, \text{ which can be scaled to}$$

$$0.1208332, 0.5816233, 0.2593327, 0.2592972, \ldots$$

The quality of a linear congruential generator can be evaluated using the Chebyshev distance between the target probability density function f_u and the generated probability density function $\hat{f}$, which is defined as

$$D(f_u, \hat{f}) = \max_{u \in \mathbb{R}} |f_u(u) - \hat{f}(u)|$$

Ideally, a well-performing linear congruent generator should achieve consistent convergence, which means that, for any data sequence, the maximum loss approaches 0 as the sequence grows infinitely large, meaning that

$$\max_{\text{all data}} D(f_u, \hat{f}) \xrightarrow{n \to +\infty} 0$$

Most of the time, however, it is sufficient to have the average error to vanish for a linear congruential generator to be good.

6.2.1 The period of a linear congruential generator

For certain choices of α and β, a linear congruential generator has a period equal to M and, when scaled by $(M + 1)$, it can act as a generator on $(0, 1)$.
More specifically, a linear congruential generator has a period equal to M for any starting seed u_0 if and only if:

- M and α are co-prime, which means that $\text{GCD}(M, \alpha) = 1$.
- $(\beta - 1)$ is divisible by all prime factors of M.
- $(\beta - 1)$ is a multiple of 4 if M is a multiple of 4.

6.2.2 The distribution of a linear congruential generator

Generally speaking, it is possible to assess whether the generated random variable produces a sequence resembling a $Unif(0,1)$ random sample through dedicated statistical tests.

6.2.3 Simulating independence in a linear congruential generator

Since each generated sample u_n is a deterministic function of the previous value u_{n-1}, linear congruential generators tend to create dependent sequences.
To simulate independent samples, a linear congruential generator must be k-dimensionally equidistributed, with k being chosen as a large value, which means that every subsequence of length k occurs with equal frequency.

6.3 Transformations of random variables

It is possible to simulate (continuous) random variables through specific transformations.

6.3.1 Using quantile transforms to generate populations

The quantile transform can be used to create random variables starting from a $Unif(0,1)$ generator.

Example: Generating a continuous random variable

Let X be a continuous random variable for which

$$f_X(x) = 3x^2 \mathbb{I}_{[0,1]}(x) \text{ and } F_X(x) = \begin{cases} 0 & \text{if } x < 0 \\ x^3 & \text{if } x \in [0,1) \\ 1 & \text{if } x \geq 1 \end{cases}$$

Since F_X is an invertible function, it is possible to define the quantile function $\mathbb{Q}_X(p) = \sqrt[3]{p}$, with $p \in (0,1)$.
Therefore, if $U \sim Unif(0,1)$, the random variable $\sqrt[3]{U}$ follows the same distribution as X.

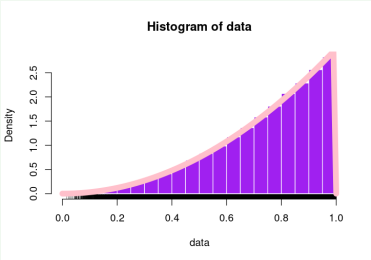


Figure 6.1: The distribution of $\sqrt[3]{U}$, with $U \sim Unif(0,1)$.

While easy to implement, this algorithm becomes more cumbersome when generating discrete random variables.

6.3.2 Deterministic transformations of random variables

Starting from a continuous random variable X that, on a support set S_X, has probability density function $f_X(x)$ and cumulative distribution $F_X(x)$, it is possible to study the random variable $Y = r(X)$ by defining the set $A_y = \{x \in S_X : r(x) \leq y\}$ and using it to compute

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(r(X) \leq y) = \mathbb{P}(\{x \in S_X : r(x) \leq y\}) = \int_{A_y} f_X(x)\ dx$$

$$f_Y(y) = F'_Y(y)\mathbb{I}_{S_Y}(y), \text{ with } S_Y = \{r(x) : x \in S_X\}.$$

However, if the function $r(x)$ is known to be invertible in S_X, with $s(y) = r^{-1}(x)$, it is possible to directly compute

$$F_Y(y) = \mathbb{P}(r(X) \leq y) = \mathbb{P}(X \leq s(y)) = F_X(s(y))$$
$$f_Y(y) = F'_Y(y) = f_X(s(y)) \cdot |s'(y)|$$

Example: Transforming a uniformly distributed random variable

Given a random variable $X \sim Unif(-1, 1)$, define the random variable $Y = X^2$, whose cumulative distribution function is defined as

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(X^2 \leq y) = \mathbb{P}(-\sqrt{y} \leq X \leq \sqrt{y}) = \int_{-\sqrt{y}}^{\sqrt{y}} \frac{1}{2}\ dx = \frac{x}{2}\Big|_{-\sqrt{y}}^{\sqrt{y}} = \sqrt{y}$$

This means that, on the support $S_Y = \{x^2 : x \in S_X\} = [0, 1]$, the probability density function of Y will be given by

$$f_Y(y) = F'_y(y) = \frac{1}{2\sqrt{y}}\mathbb{I}_{(0,1]}(y)$$

This concept can be generalized to create transformations of random vectors.
For a couple of random variables X and Y with joint probability density function $f_{X,Y}(x, y)$, it is possible to study the many-to-one (deterministic) transformation $Z = r(X, Y)$ by defining the set $A_z = \{(x, y) \in S_X \times S_Y : r(x, y) \leq z\}$ and using it to compute

$$F_Z(z) = \mathbb{P}(Z \leq z) = \mathbb{P}(r(X, Y) \leq z) = \mathbb{P}(\{(x, y) \in S_X \times S_Y : r(x, y) \leq z\}) = \int\int_{A_z} f_{X,Y}(x, y)\ dxdy$$

$$f_Z(z) = F'_Z(z)$$

This strategy can be extended to certain many-to-many (deterministic) transformations as well.
From a broader perspective, let $\mathbf{r} : \mathbb{R}^p \to \mathbb{R}^q$ be a differentiable vector field such that each component $r_1, \ldots, r_q$ is a function $r_i : \mathbb{R}^p \to \mathbb{R}$: the Jacobian matrix $\nabla_{\mathbf{r}}$ is a $(q \times p)$ matrix that contains all the partial derivatives of $\mathbf{r}$, meaning that

$$\nabla_{\mathbf{r}}[i, j] = \frac{\partial r_i}{\partial x_j}\ \forall (i, j) \in \{1, \ldots, q\} \times \{1, \ldots, p\}$$

Therefore, given two open sets $A, B \subseteq \mathbb{R}^p$, consider a random vector $\mathbf{X}$ that, on the set A, has joint probability density function $f_{\mathbf{X}}(\mathbf{x})$ and define $\mathbf{r} : A \to B$ as a (deterministic) bijective and

differentiable vector field with continuous partial derivatives: the random vector $\mathbf{Y} = \mathbf{r}(\mathbf{X})$ can be studied by inverting its defining system of equations and considering

$$\mathbf{y} = \mathbf{r}(\mathbf{x}) \Leftrightarrow \begin{cases} y_1 = r_1(\mathbf{x}) \\ y_2 = r_2(\mathbf{x}) \\ \vdots \\ y_p = r_p(\mathbf{x}) \end{cases} \rightarrow \begin{cases} x_1 = s_1(\mathbf{y}) \\ x_2 = s_2(\mathbf{y}) \\ \vdots \\ x_p = s_p(\mathbf{y}) \end{cases} \Leftrightarrow \mathbf{x} = \mathbf{s}(\mathbf{y})$$

$$f_{\mathbf{Y}}(\mathbf{y}) = f_{\mathbf{X}}(\mathbf{s}(\mathbf{y})) \cdot |det(\nabla_{\mathbf{s}}(\mathbf{y}))| \cdot \mathbb{I}_B(\mathbf{y})$$

Example: Creating a t distribution using a deterministic transformation

Consider two independent random variables $Z \sim \mathcal{N}(0,1)$ and $W \sim \chi^2_\nu$: it is possible to show that

$$T_\nu = \frac{Z}{\sqrt{\frac{W}{\nu}}} \sim t_\nu \Rightarrow f_T(t) \propto \left(1 + \frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}$$

Since a many-to-one transformation is not bijective, it is possible to introduce a new variable $X \sim \chi^2_\nu$ so that $f_{T_\nu}(t)$ can be retrieved from the joint probability density function $f_{T_\nu,X}(t,x)$. For this reason, define the following many-to-many transformation:

$$\begin{pmatrix} T_\nu \\ X \end{pmatrix} = \mathbf{r}(Z,W) = \begin{pmatrix} \frac{Z}{\sqrt{\frac{W}{\nu}}} \\ W \end{pmatrix} \Leftrightarrow \begin{pmatrix} Z \\ W \end{pmatrix} = \mathbf{s}(T_\nu, X) = \begin{pmatrix} T_\nu \sqrt{\frac{X}{\nu}} \\ X \end{pmatrix}$$

Its Jacobian matrix will be given by

$$\nabla_{\mathbf{s}} = \begin{pmatrix} \frac{\partial s_1}{\partial t} & \frac{\partial s_1}{\partial x} \\ \frac{\partial s_2}{\partial t} & \frac{\partial s_2}{\partial x} \end{pmatrix} = \begin{pmatrix} \sqrt{\frac{x}{\nu}} & \frac{t}{2\sqrt{\nu x}} \\ 0 & 1 \end{pmatrix} \Rightarrow |det(\nabla_{\mathbf{s}})| = \sqrt{\frac{x}{\nu}}$$

Therefore, it is possible to find the joint probability density function of T_ν and X by computing

$$f_{T_\nu,X}(t,x) = f_{Z,W}(\mathbf{s}(t,x)) \cdot |det(\nabla_{\mathbf{s}})| \overset{Z \perp W}{=} f_Z\left(t\sqrt{\frac{x}{\nu}}\right) f_W(x) \cdot \sqrt{\frac{x}{\nu}}$$

$$\propto e^{-\frac{1}{2}(t^2\frac{x}{\nu})} x^{\frac{\nu}{2}-1} e^{-\frac{\nu}{2}} \sqrt{\frac{x}{\nu}} = e^{-\frac{1}{2}x(1+\frac{t^2}{\nu})} x^{\frac{\nu+1}{2}-1}$$

The marginal distribution of T_ν will ultimately be given by

$$f_{T_\nu}(t) = \int_0^{+\infty} f_{T_\nu,X}(t,x)\, dx \propto \int_0^{+\infty} x^{\frac{\nu+1}{2}-1} e^{-\frac{1}{2}(1+\frac{t^2}{\nu})x}\, dx \xrightarrow[\text{Gamma integral}]{u=\frac{1}{2}(1+\frac{t^2}{\nu})}$$

$$f_{T_\nu}(t) \propto \frac{\Gamma(\frac{\nu+1}{2})}{(1+\frac{t^2}{\nu})^{\frac{\nu+1}{2}}} \propto \left(1 + \frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}$$

For instance, the Box-Muller transform is a many-to-many transformation that allows to convert

pairs of uniformly distributed samples into pairs of normally distributed samples, starting from two independent random variables $X, Y \sim \mathcal{N}(0,1)$ whose joint probability density function is

$$f_{X,Y}(x,y) = \frac{1}{2\pi} e^{-\frac{x^2+y^2}{2}}$$

This function can be simplified through an inverse transformation based on polar coordinates:

$$\begin{pmatrix} X \\ Y \end{pmatrix} = \mathbf{s}(R,\Theta) = \begin{pmatrix} R\cos(\Theta) \\ R\sin(\Theta) \end{pmatrix} \Leftrightarrow \begin{pmatrix} R \\ \Theta \end{pmatrix} = \mathbf{r}(X,Y) = \begin{pmatrix} \sqrt{X^2+Y^2} \\ atan2(Y,X) \end{pmatrix}$$

In fact, this transformation is invertible in the open set $\mathcal{D} = (0,+\infty) \times (0,2\pi)$, where the Jacobian matrix is given by

$$\nabla_{\mathbf{s}} = \begin{pmatrix} \frac{\partial s_1}{\partial r} & \frac{\partial s_1}{\partial \theta} \\ \frac{\partial s_2}{\partial r} & \frac{\partial s_2}{\partial \theta} \end{pmatrix} = \begin{pmatrix} \cos(\theta) & -r\sin(\theta) \\ \sin(\theta) & r\cos(\theta) \end{pmatrix} \Rightarrow |det(\nabla_{\mathbf{s}})| = r$$

Therefore, it is possible to reformulate the joint probability density function by considering

$$\begin{aligned} f_{R,\Theta}(r,\theta) = f_{X,Y}(\mathbf{s}(r,\theta)) \cdot |det(\nabla_{\mathbf{s}})| \cdot \mathbb{I}_{\mathcal{D}}(r,\theta) &= \frac{1}{2} r e^{-\frac{1}{2}r^2} \mathbb{I}_{(0,+\infty)\times(0,2\pi)}(r,\theta) \\ &= \left(r e^{-\frac{1}{2}r^2} \mathbb{I}_{\mathbb{R}^+}(r) \right) \left(\frac{1}{2\pi} \mathbb{I}_{(0,2\pi)}(\theta) \right) = f_R(r) f_\Theta(\theta) \end{aligned}$$

Knowing that $f_\Theta(\theta)$ allows to draw samples from a $Unif(0,2\pi)$ distribution, it is possible to construct samples from $f_R(r)$ using the quantile transform method.
The first step consists of retrieving the cumulative distribution function of R:

$$F_R(r) = \mathbb{P}(R \leq r) = \int_0^r f_R(t)\, dt = \int_0^r t e^{-\frac{1}{2}t^2}\, dt = -e^{-\frac{t^2}{2}} \Big|_0^r = \left(1 - e^{-\frac{1}{2}r^2}\right) \mathbb{I}_{\mathbb{R}^+}(r)$$

Since $F_R(r)$ is an invertible function in $\mathbb{R}^+$, its quantile function can be computed directly as

$$p = F_R(r) = 1 - e^{-\frac{1}{2}r^2} \Leftrightarrow r = \sqrt{-2\log(1-p)} = \mathbb{Q}(p)$$

Therefore, it is possible to draw samples from $R = \mathbb{Q}(U) = \sqrt{-2\log(U)}$, where $U \sim Unif(0,1)$ and, consequently, $(1-U) \sim Unif(0,1)$ as well.

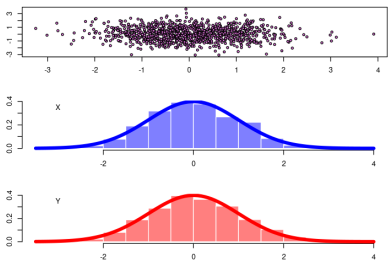


Figure 6.2: Visualization of the Box-Muller transform.

Chapter 7

Graphical models

7.1 The Markov assumption

Generally speaking, the joint distribution $p_{\mathbf{X}}(\mathbf{x})$ of a discrete random vector $\mathbf{x} \in \mathbb{R}^D$ can be described using the chain rule, resulting in

$$p_{\mathbf{X}}(\mathbf{x}) = p_{X_1,\dots,X_D}(x_1,\dots,x_D) = \mathbb{P}(x_1)\mathbb{P}(x_2|x_1)\mathbb{P}(x_3|x_1,x_2)\dots\mathbb{P}(x_D|x_1,\dots,x_{D-1})$$

However, as the number of components in the random vector increases, modelling this distribution becomes computationally and statistically unviable.
In fact, assuming each variable in the random vector can take J possible values, the conditional probability table that describes the factor $\mathbb{P}(x_d|x_1,\dots,x_{d-1})$ requires $O(J^d)$ values, meaning that a total of $O(J^D)$ parameters would be needed to represent the full joint distribution.
However, it is possible to simplify this joint distribution by introducing some assumptions about the conditional independence between random variables.
From a broader perspective, two random variables X and Y are said to be conditionally independent given the random variable Z if and only if

$$\mathbb{P}(X,Y|Z) = \mathbb{P}(X|Z)\mathbb{P}(Y|Z) \text{ or } \mathbb{P}(X|Y,Z) = \mathbb{P}(X|Z)$$

This concept allows to formulate the first-order Markov assumption, which states that "the future is independent of the past given the present", as follows:

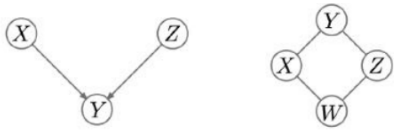
$$p_{\mathbf{X}}(\mathbf{x}) = \mathbb{P}(x_1) \cdot \prod_{d=2}^{D} \mathbb{P}(x_d|x_{d-1})$$

Overall, this formulation is computationally lighter, although it only works for one-dimensional data sequences, whereas higher-dimensional data sequences require more advanced tools, such as graphical models.

7.2 Formalizing (directed) graphical models

A graphical model allows to represent conditional independence assumptions using a graph $\mathcal{G} = (V, E)$ that is described by a node set V, which represents the components of a random vector, and an

edge set E, which indicates the conditional independence relations.



Two graphs whose conditional independence properties can not be perfectly expressed by a graph in the other family.

Figure 7.1: Examples of directed and undirected graphical models.

Most graphical models are based on directed acyclic graphs because they allow for a topological ordering, ensuring that all parent nodes come before their child nodes.

Directed Graphical Models

Graph Terminology

Parent: The parents of a node is the set of all nodes that feed into it.

Child: The children of a node is the set of all nodes that feed out of it.

Root: A root is a node with no parents.

Leaf: A leaf is a node with no children.

Ancestors: The ancestors are the parents, grand–parents, etc of a node.

Descendants: The descendants are the children, grand-children, etc of a node.

Neighbors: The neighbors of a node is the set of all immediately connected nodes.

Degree: The degree of a node is the number of neighbors. For directed graphs, we speak of the *in–degree* and *out–degree*, which count the number of parents and children.

Cycle or Loop: A cycle is a series of nodes such that we can get back to where we started by following edges. We may speak of a directed or undirected cycles.

Path: A path $s \rightsquigarrow t$ is a series of directed edges leading from s to t.

Tree: An *undirected tree* is an undirected graph with no cycles. A *directed tree* is a DAG in which there are no directed cycles.

Forest: A forest is a set of trees.

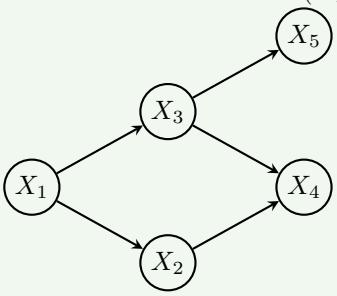
Figure 7.2: Graph terminology for directed graphical models.

In fact, for a directed graph $\mathcal{G} = (V, E)$, where $V = \{X_1, \ldots, X_D\}$, the ordered Markov property denotes the assumption that "each node in the graph depends only on its immediate parents", which can be formulated as

$$(X_d \perp \mathbf{X}_{predecessors(d) \setminus parents(d)} | \mathbf{X}_{parents(d)}) \ \forall d \in \{1, \ldots, D\}$$

Example: Factorizing a joint distribution using a graphical model

For a random vector $\mathbf{X} = (X_1, X_2, X_3, X_4, X_5)$, consider the following directed acyclic graph:



Using the information provided by the graph, the joint distribution can be simplified to

$$p_{\mathbf{X}}(\mathbf{x}) \overset{\text{Chain rule}}{=} \mathbb{P}(x_1)\mathbb{P}(x_2|x_1)\mathbb{P}(x_3|x_1,x_2)\mathbb{P}(x_4|x_1,x_2,x_3)\mathbb{P}(x_5|x_1,x_2,x_3,x_4)$$
$$\overset{\text{DAG}}{=} \mathbb{P}(x_1)\mathbb{P}(x_2|x_1)\mathbb{P}(x_3|x_1)\mathbb{P}(x_4|x_2,x_3)\mathbb{P}(x_5|x_3)$$

In a more general sense, the ordered Markov property results in

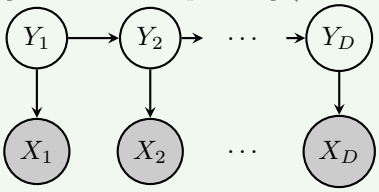
$$p_{\mathbf{X}}(\mathbf{x}|\mathcal{G}) = \prod_{d=1}^{D} \mathbb{P}(x_d|\mathbf{x}_{parents(d)}), \text{ provided each assumption is correct and satisfied.}$$

In particular, the set $\mathcal{M}(\mathcal{G})$ contains all the joint distributions $\mathbb{P}$ that, by satisfying the Markov property, are said to be "Markov to $\mathcal{G}$".
However, this set can also contain distributions that are "not faithful to $\mathcal{G}$", meaning that they define additional independence relations that are not captured by $\mathcal{G}$, requiring to consider

$$\mathcal{M}(\mathcal{G}) = \{\mathbb{P} \text{ is a joint distribution over } \mathbf{X} : CI(\mathcal{G}) \subseteq CI(\mathbb{P})\}$$

Example: Hidden Markov models

A first-degree Hidden Markov model is defined using a sequence of marginally dependent (discrete) random variables $X_1, \ldots, X_D$ such that each variable X_i is conditionally independent given the corresponding (discrete) latent variable Y_i.

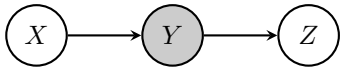


Looking at the structure of its graph, the joint probability distribution of a Hidden Markov model can be expressed as

$$p_{\mathbf{X},\mathbf{Y}}(\mathbf{x},\mathbf{y}) = \mathbb{P}(y_1)\left[\prod_{d=1}^{D}\mathbb{P}(x_d|y_d)\right]\left[\prod_{t=2}^{D}\mathbb{P}(y_t|y_{t-1})\right]$$

Overall, conditional independence in directed acyclic graphs can be determined by checking whether two sets of random variables are connected conditioned on a third set of variables, with connection indicating conditional dependence and separation implying conditional independence.
For a joint distribution $\mathbb{P} \in \mathcal{M}(\mathcal{G})$, a generic conditional independence statement $(\mathbf{X}_A \perp \mathbf{X}_B | \mathbf{X}_C)$ holds if every variable in the set A is separated from every variable in the set B, conditioned on all variables in the set C.
Rather than checking every single path between a variable in the set A and a variable in the set B, it is sufficient to consider triples of nodes X, Y and Z that form one of the following structures:

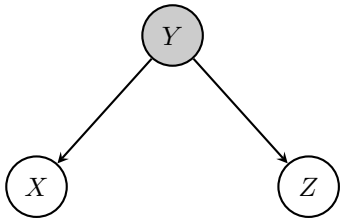
1. **Chain:** This structure substantially represents the first-order Markov assumption at the base of (directed) graphical models.



   Indeed, the structure of this graph reveals that

$$\mathbb{P}(x, y, z) = \mathbb{P}(x)\mathbb{P}(y|x)\mathbb{P}(z|y)$$
$$\mathbb{P}(z|x, y) = \frac{\mathbb{P}(x, y, z)}{\mathbb{P}(x, y)} = \frac{\mathbb{P}(x)\mathbb{P}(y|x)\mathbb{P}(z|y)}{\mathbb{P}(x)\mathbb{P}(y|x)} = \mathbb{P}(z|y) \Rightarrow (X \perp Z|Y)$$

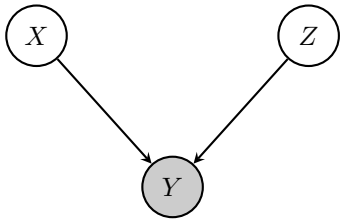
2. **Common cause:** This structure indicates that Y acts as a cause for both X and Z.



   In particular, the structure of this graph reveals that

$$\mathbb{P}(x, y, z) = \mathbb{P}(y)\mathbb{P}(x|y)\mathbb{P}(z|y)$$
$$\mathbb{P}(x, z|y) = \frac{\mathbb{P}(x, y, z)}{\mathbb{P}(y)} = \frac{\mathbb{P}(y)\mathbb{P}(x|y)\mathbb{P}(z|y)}{\mathbb{P}(y)} = \mathbb{P}(x|y)\mathbb{P}(z|y) \Rightarrow (X \perp Z|Y)$$

3. **Colliders:** This structure indicates that both X and Z can act as causes for Y.



   An interesting property related to this structure is Berkson's paradox, which states that, in this scenario, X and Z are marginally independent, but become conditionally dependent given

Y because one would have

$$\mathbb{P}(x,y,z) = \mathbb{P}(x)\mathbb{P}(z)\mathbb{P}(y|x,z)$$

$$\mathbb{P}(x,z) = \sum_{y} \mathbb{P}(x,y,z) = \mathbb{P}(x)\mathbb{P}(z)\left(\sum_{y} \mathbb{P}(y|x,z)\right) = \mathbb{P}(x)\mathbb{P}(z) \Rightarrow X \perp Z$$

$$\mathbb{P}(x,z|y) = \frac{\mathbb{P}(x,y,z)}{\mathbb{P}(y)} = \frac{\mathbb{P}(x)\mathbb{P}(z)\mathbb{P}(y|x,z)}{\mathbb{P}(y)} \neq \mathbb{P}(x|y)\mathbb{P}(z|y) \Rightarrow (X \not\perp Z|Y)$$

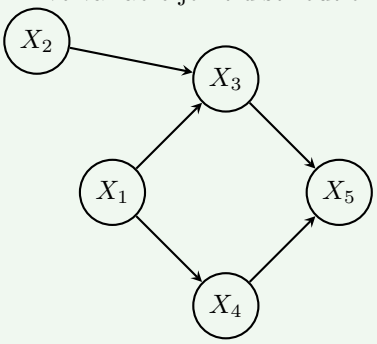
Using this information, it is possible to define an algorithm that, given a directed graph $\mathcal{G} = (V, E)$ and three disjoint sets $\mathbf{X}_A$, $\mathbf{X}_B$ and $\mathbf{X}_C$, checks whether the graph implies that $(\mathbf{X}_A \perp \mathbf{X}_B | \mathbf{X}_C)$. This algorithm considers all possible undirected paths $\mathcal{U}$ from any node in the set $\mathbf{X}_A$ to any node in the set $\mathbf{X}_B$ and states that a path is "blocked" if and only if at least one of the following conditions is met:

1. $\mathcal{U}$ contains a chain that passes through some node in the set $\mathbf{X}_C$.
2. $\mathcal{U}$ contains a common clause that starts from a node in the set $\mathbf{X}_C$.
3. $\mathcal{U}$ contains a collider that converges to a node that is not in the set $\mathbf{X}_C$ and whose descendants are not in the set $\mathbf{X}_C$ either.

Therefore, if all the checked paths are blocked, it is possible to conclude that $\mathbf{X}_A$ is separated from $\mathbf{X}_B$ by $\mathbf{X}_C$ and the joint distribution represented by the graph satisfies the conditional independence assumption that $(\mathbf{X}_A \perp \mathbf{X}_B | \mathbf{X}_C)$.

Example: An application of the separation algorithm

A five-variable joint distribution is Markov to the following graph:



The structure of the graph allows to show that $(\{X_4, X_5\} \perp X_2 | \{X_1, X_3\})$:

1. The path $X_2 \rightarrow X_3 \rightarrow X_5 \leftarrow X_4$ is blocked because $X_5 \notin \{X_1, X_3\}$ acts as a collider for the graph.
2. The path $X_2 \rightarrow X_3 \rightarrow X_5$ is blocked because it represents a chain passing through $X_3 \in \{X_1, X_3\}$.

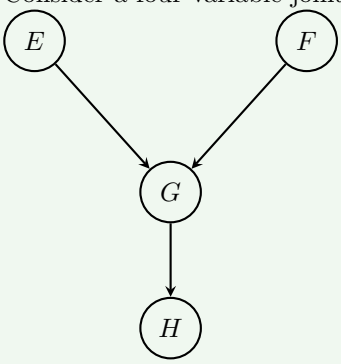
Since both paths are blocked, the assumption that $(\{X_4, X_5\} \perp X_2 | \{X_1, X_3\})$ is indeed true.

7.3 Using directed acyclic graphs for ancestral sampling

Ancestral sampling is a generative technique that exploits the Markov property to draw samples from a joint distribution encoded by a directed acyclic graph.

Example: Modelling data distribution from a graphical model

Consider a four-variable joint distribution that is represented by the following graph:



Using the Markov property, this joint distribution can be expressed as

$$\mathbb{P}(e, f, g, h) = \mathbb{P}(e)\mathbb{P}(f)\mathbb{P}(g|e, f)\mathbb{P}(h|g)$$

Assuming $(H|G = g) \sim Exp(g)$ and $(G|E = e, F = f) \sim Gamma(e, f)$, it is necessary to set some constraints on the support of each variable.
Since g is a parameter for an exponential distribution, its support must be a subset of $\mathbb{R}^+$.
Similarly, knowing that e and f are used to model a Gamma distribution, their supports must also lie within $\mathbb{R}^+$.
For this reason, the marginal distributions of E and F must be uniform, provided the support is defined as a subset of $\mathbb{R}^+$, or belong to the Gamma or exponential families.
Therefore, drawing a single sample from the joint distribution requires (independently) sampling e and f from their respective distributions and using them to draw g, which, in turn, allows to sample h.

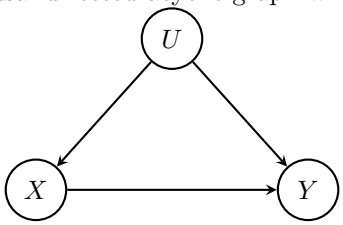
7.4 Causal inference

Causation occurs whenever a change in one variable directly produces a change in another variable. However, an association, which describes a statistical relationship or dependence between variables, does not necessarily indicate a cause-and-effect relationship because it may arise from a "spurious correlation" in which the variables are actually influenced by a hidden "confounding" variable.

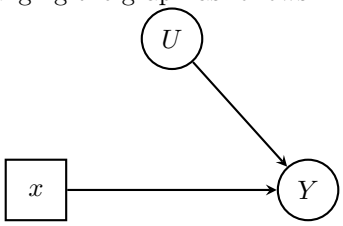
Example: Conspiracy theories in the era of COVID-19

In the early 2020s, conspiracies about the COVID-19 pandemic arose because many reported cases in the United States of America occurred in the proximity of 5G cable stations. However, this relationship can be explained by considering the number of people living in these regions as a larger population introduces a higher risk of spreading the disease while also providing an incentive to build cable stations nearby.

Causation can be formally described using causal directed acyclic graphs.
Assuming X is an observable potential cause and Y is an observed effect, define the random vector $\mathbf{U}$ as a collection of (unobservable) confounding variables that can affect X or Y: the resulting causal directed acyclic graph will be structured as



By the Markov property, this graph models the joint distribution $\mathbb{P}(x, y, \mathbf{u}) = \mathbb{P}(\mathbf{u})\mathbb{P}(x|\mathbf{u})\mathbb{P}(y|x, \mathbf{u})$.
More specifically, while the association between X and Y can be represented by the conditional distribution $\mathbb{P}(y|\text{Observe } X = x) = \mathbb{P}(y|x)$, the causal distribution requires fixing some $X = x$ and changing the graph as follows:



In fact, using the point mass distribution $\delta_X(x)$, the causal distribution can be retrieved by computing

$$\mathbb{P}(y|\text{Set } X = x) = \int \mathbb{P}(y|x, \mathbf{u})\mathbb{P}(\mathbf{u}) \, d\mathbf{u}$$

Overall, an association indicates a passive conditioning based on observations, whereas a causal distribution describes an active conditioning based on intervention.
However, a causal distribution cannot be estimated directly because it depends on a collection of not (entirely) observable variables, requiring to collect data in order to measure (some elements) of $\mathbf{U}$.

Chapter 8

Exploratory data analysis

8.1 Learning distributions through graphical summaries

Assuming a continuous random variable follows an unknown probability distribution, several strategies allow to learn this distribution starting from a data sample $\mathcal{D}_n = \{x_1, \ldots, x_n\}$ and plotting its behaviour.

8.1.1 Empirical cumulative distribution functions and histograms

It is possible to learn the cumulative distribution function of an unknown distribution through the empirical cumulative distribution function, which represents the cumulative distribution of a discrete distribution that assigns the same mass to each data point.
For this reason, the empirical cumulative distribution function can be defined as

$$\hat{F}_n(x) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}(x_i \leq x), \text{ where } \mathbf{1}(x_i \leq x) = \begin{cases} 1 & \text{if } x_i \leq x \\ 0 & \text{otherwise} \end{cases}$$

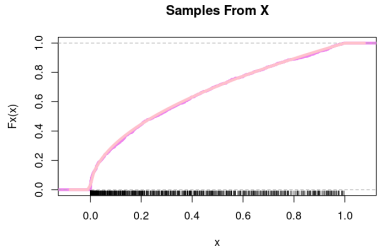


Figure 8.1: Learning a distribution using the empirical cumulative distribution function.

Alternatively, a model can learn the probability density function of the distribution using a histogram. In fact, on a support S_X, a histogram makes use of n bins $\{B_1, \ldots, B_n\}$, each with a fixed size h, to

estimate the probability density function within each bin as

$$\hat{f}_X(x) = \frac{1}{n|B_j|} \sum_{i=1}^{n} \mathbf{1}(x_i \in B_j), \text{ where } \mathbf{1}(x_i \in B_j) = \begin{cases} 1 & \text{if } x_i \in B_j \\ 0 & \text{otherwise} \end{cases}$$

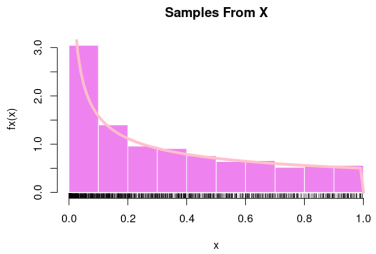


Figure 8.2: Learning a distribution using a histogram.

While histograms can be generalized to learn multivariate distributions, their usage remains more limited because they provide good results only when the probability density function is assumed to be homogeneously smooth.

8.1.2 Kernel density estimation

One of the main issues of learning continuous distributions using an empirical cumulative distribution function or a histogram lies in the fact that a step function is used to represent a smooth probability density function.
For this reason, graphical exploratory data analysis is often conducted using kernel density estimation, which consists of applying a smoothing kernel K that distributes the weight of each data point within a neighbourhood of that data point.
Therefore, the estimated probability density function will be given by

$$\hat{f}_X(x) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right), \text{ where } h \text{ is the bandwidth of the chosen kernel.}$$

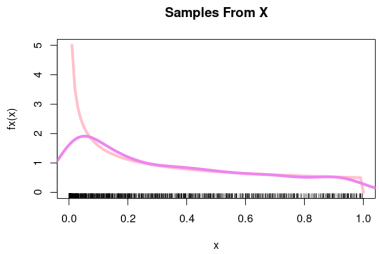


Figure 8.3: Learning a distribution using kernel density estimation.

However, it should be noticed that, when performed on a distribution with a bounded support, kernel density estimation often results in the loss of information as some mass is placed outside of the support of the distribution.